

Sleep Study - Amy Bohnert - 2/27/2025

```
#Description of the models is here:
#https://docs.google.com/document/d/1wPB0gAGN5HowoabCOF4LSJFSpuju6Igh/edit

#Understanding geoids
#https://www.census.gov/programs-surveys/geography/guidance/geo-identifiers.html

library(foreign)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(lattice)
library(lcsm)
```

This is lcsm 0.3.2
Please report any issues or ideas at:
<https://github.com/milanwiedemann/lcsm/issues>

```
library(ggplot2)
library(tidyr)
library(dplyr)
```

```
library(stringr)
library(lmerTest)
```

Loading required package: lme4
Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Attaching package: 'lmerTest'

The following object is masked from 'package:lme4':

lmer

The following object is masked from 'package:stats':

step

```
library(sjstats)
library(MuMIn)
```

```
#ID 2561 was on two rows. It seems very clear that this should only have been one row.
dat <- read.csv("../bohnert_sleep/CL_OTR_COI_Data.csv")
dat2 <- read.csv("../bohnert_sleep/CL_OTR v1 Youth Baseline COI3 Actigraphy Demo.csv")
```

Data Cleaning and EDA

- Observation ID 2561 had two rows in the data set. I removed one of these rows.

```
#EDA
#ID 2561 is a problem
#dat %>% group_by(ID) %>% summarize(n = n()) %>% arrange(-n)
dat %>% filter(ID == 2561) %>% select(RecordedDate, ID, age.v1, gender.v1)
```

	RecordedDate	ID	age.v1	gender.v1
1	1/30/2020 12:50	2561	NA	NA
2	<NA>	2561	19	2

```
#Remove ID's that are NA
dat <- dat %>% filter(!is.na(ID))

#ID 2561 has two rows
#Fixing this here:
dat[dat$ID == 2561 &
      is.na(dat$age.v1), c("age.v1", "age_dichotomous", "gender.v1")] <-
  dat[dat$ID == 2561 & !is.na(dat$age.v1), c("age.v1", "age_dichotomous", "gender.v1")]
dat <- dat[dat$ID != 2561 |
            (dat$ID == 2561 & !is.na(dat$RecordedDate)), ]

#Convert gender 1/2 to male/female
dat <- dat %>% mutate(
  gender.v1 = case_when(gender.v1 == 1 ~ "male", gender.v1 == 2 ~ "female"),
  pdsMF.v1 = case_when(pdsMF.v1 == 1 ~ "male", pdsMF.v1 == 2 ~ "female")
)
```

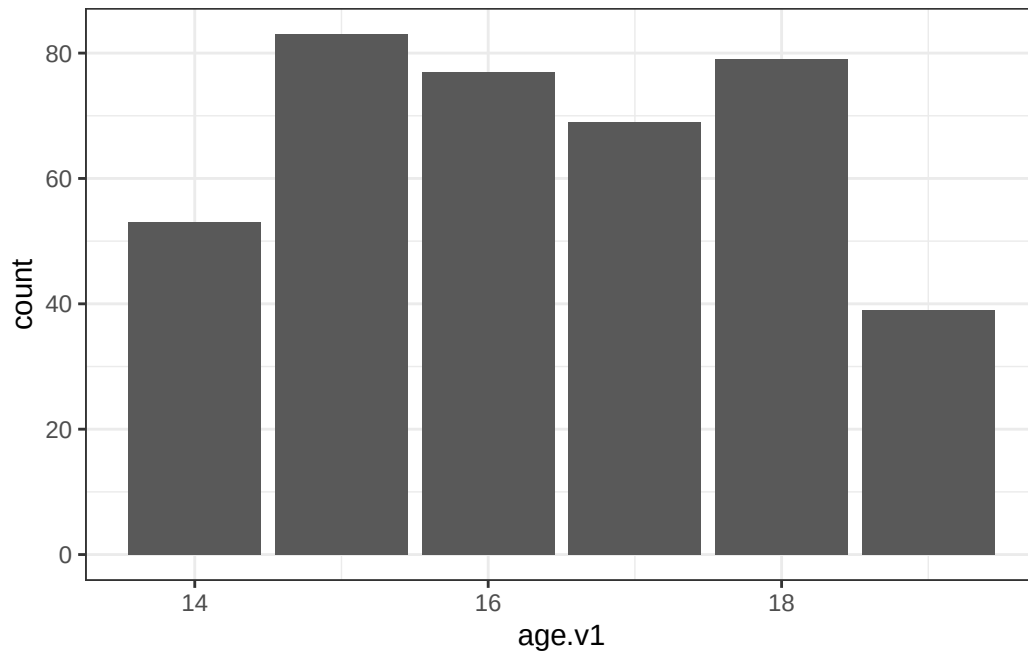
Summary stats

```
dat %>% summarize(
  mean_age = mean(age.v1, na.rm = T),
  pct_female = mean(pdsMF.v1 == "female", na.rm = T),
  mean_income = mean(income.v1, na.rm = T)
)
```

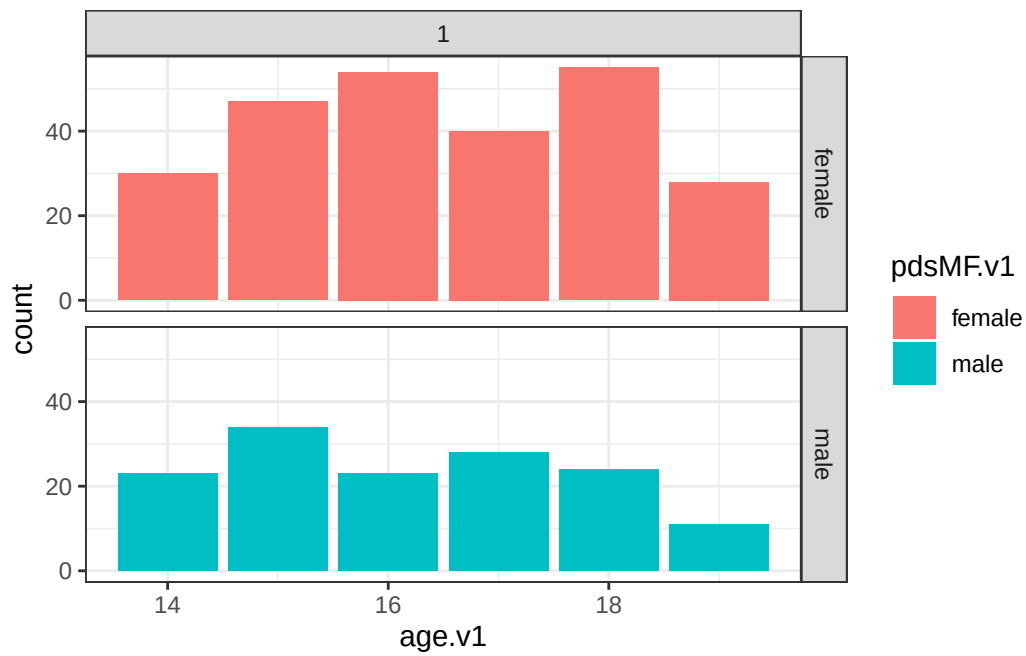
	mean_age	pct_female	mean_income
1	16.3875	0.6397985	40774.34

```
#ggplots for age and income
ggplot(aes(x = age.v1), data = dat) + geom_bar() + theme_bw()
```

Warning: Removed 3 rows containing non-finite outside the scale range (`stat\_count()`).



```
ggplot(aes(x = age.v1, fill = pdsMF.v1), data = dat %>% filter(pdsMF.v1 %in% c("male", "female"))
```



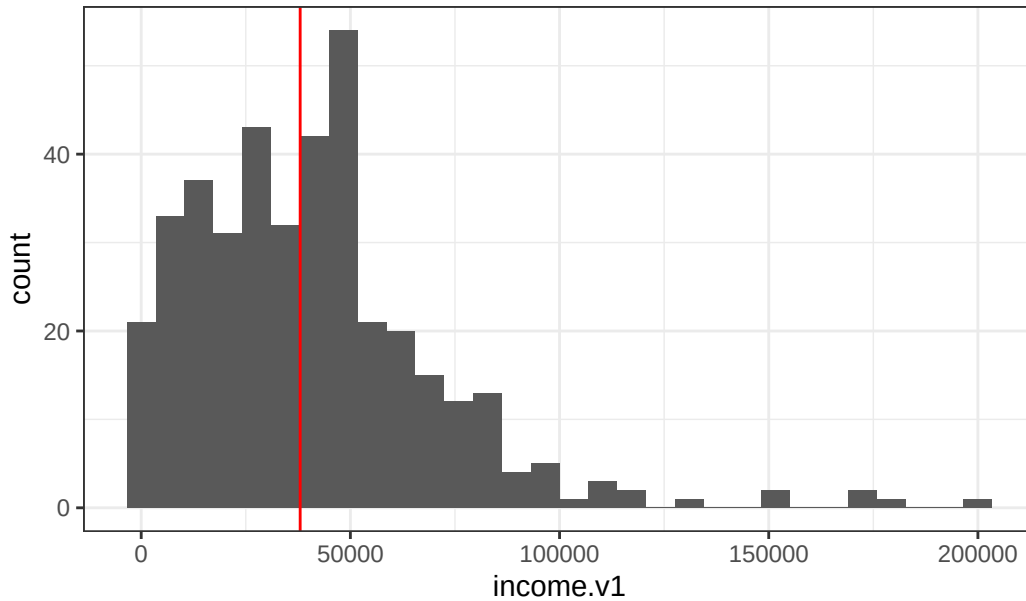
```
ggplot(aes(x = income.v1), data = dat) + geom_histogram() + theme_bw() + geom_vline(aes(xint = income.v1
```

Warning: Use of `dat\$income` is discouraged.
i Use `income` instead.

`stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.

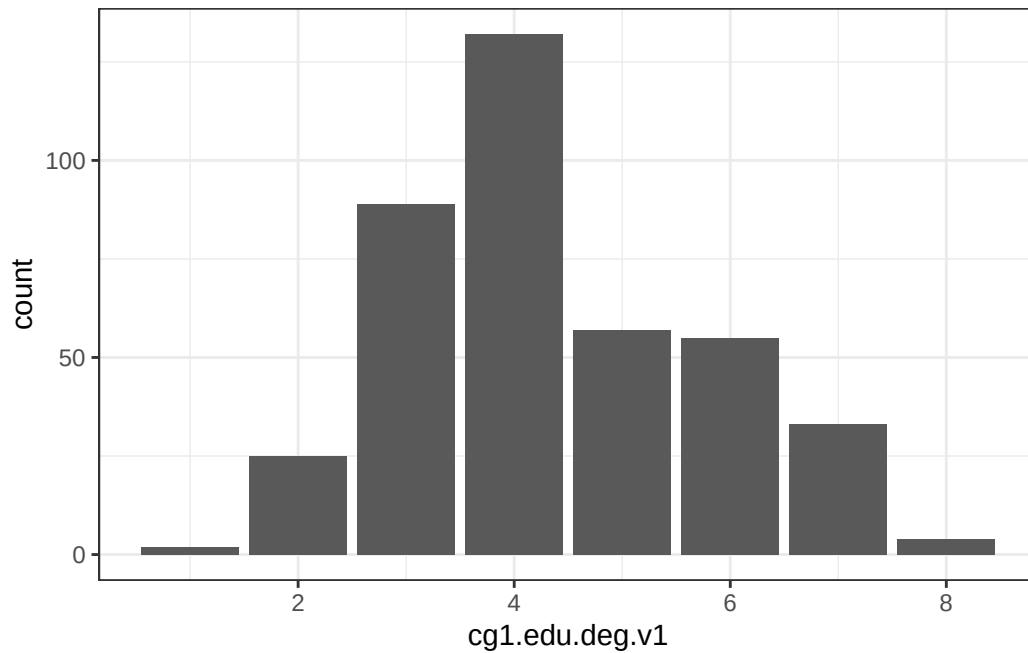
Warning: Removed 7 rows containing non-finite outside the scale range
(`stat\_bin()`).

Red line = Median

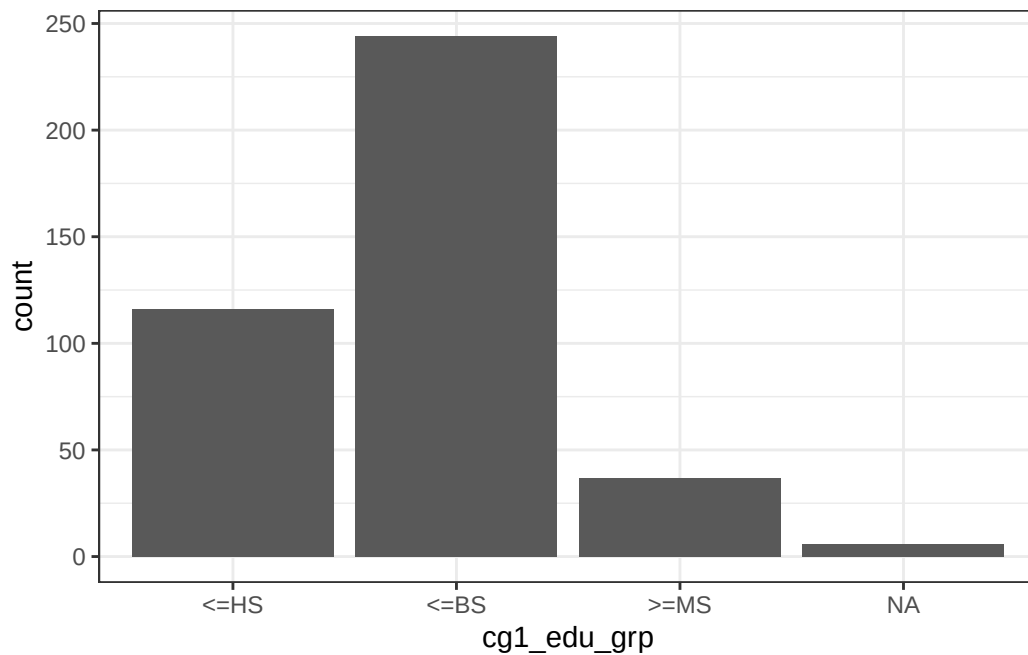


```
ggplot(aes(x = cg1.edu.deg.v1), data = dat) + geom_bar() + theme_bw()
```

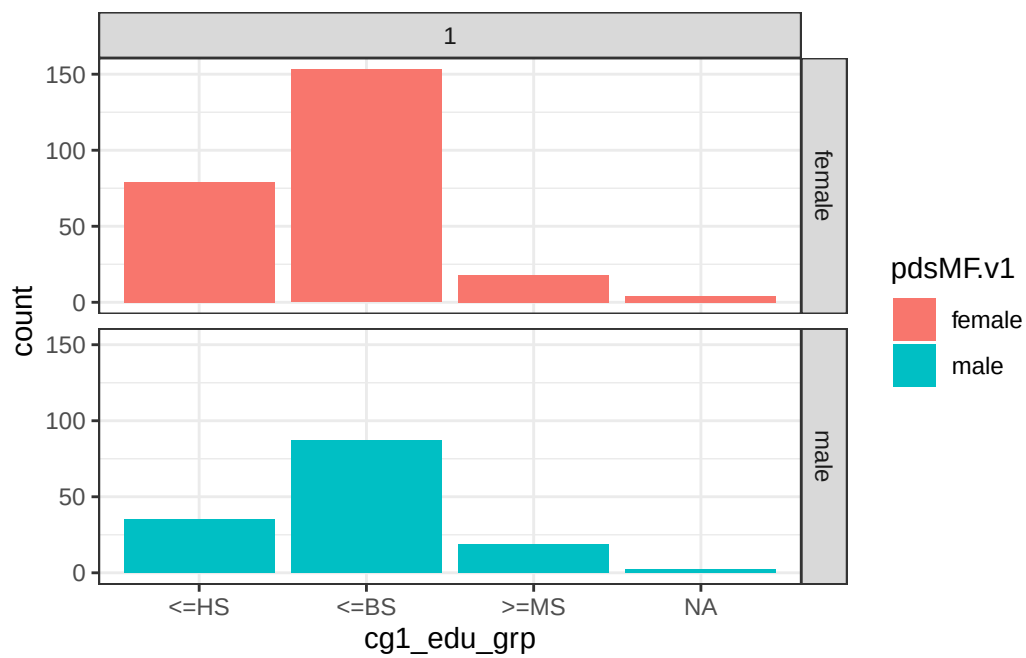
Warning: Removed 6 rows containing non-finite outside the scale range
(`stat\_count()`).



```
dat <- dat %>% mutate(cg1_edu_grp = factor(  
  case_when(  
    cg1.edu.deg.v1 %in% c(1, 2, 3) ~ "<=HS",  
    cg1.edu.deg.v1 %in% c(4, 5, 6) ~ "<=BS",  
    cg1.edu.deg.v1 %in% c(7, 8) ~ ">=MS"  
  )  
))  
  
dat <- dat %>% mutate(cg1_edu_grp = fct_relevel(cg1_edu_grp, c("<=HS", "<=BS", ">=MS")))  
ggplot(aes(x = cg1_edu_grp), data = dat) + geom_bar() + theme_bw()
```



```
ggplot(aes(x = cg1_edu_grp, fill = pdsMF.v1), data = dat %>% filter(pdsMF.v1 %in% c("male", "female"))) +
```



```
#need at least 5 days of actigraph data
table(dat$act_patotaldays)
```

0	1	2	3	4	5	6	7
8	2	4	10	18	15	55	254

```
#254+55+15 = 324
```

```
#I get 324
```

```
dat %>% filter(act_patotaldays >= 5 ) %>% nrow()
```

```
[1] 324
```

```
#Reproducing Table 1 stuff
```

```
dat %>% filter(act_patotaldays >= 5) %>% group_by(gender.v1) %>% summarize(age = mean(age.v1)
```

```
# A tibble: 3 x 3
```

	gender.v1	age	n
	<chr>	<dbl>	<int>
1	female	16.5	213
2	male	16.2	110
3	<NA>	17	1

```
dat %>% group_by(gender.v1) %>% summarize(
  age = mean(age.v1),
  n = n(),
  sleep = mean(act_totsleep.min_avg, na.rm = T),
  inbed = mean(act_totinbed.min_avg, na.rm = T),
  onset = mean(act_onset.time.min_d0, na.rm = T),
  eff = mean(act_efficiency_avg, na.rm = T)
)
```

```
# A tibble: 3 x 7
```

	gender.v1	age	n	sleep	inbed	onset	eff
	<chr>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	female	16.5	254	381.	448.	1464.	85.5
2	male	16.2	145	357.	427.	1518.	83.8
3	<NA>	NA	4	412.	504.	1441	82.4


```

#Average inbed time is a problem because times are circular
datinbed <- dat %>%
  select(ID, gender.v1, act_inbed.time_d0:act_inbed.time_d8) %>%
  pivot_longer(
    cols = starts_with("act_inbed.time_d"),
    names_prefix = "act_inbed.time_d",
    names_to = c("day"),
    values_to = "act_inbed_time"
  ) %>% mutate(act_inbed_time_numeric = as.numeric(do.call(rbind, strsplit(act_inbed_time, "
    inbed.time_angle = 2*pi*act_inbed_time_numeric/24,
    inbed.time_angle_x = cos(inbed.time_angle) ,
    inbed.time_angle_y = sin(inbed.time_angle),
  )

datweekend_yn <- dat %>%
  select(ID, act_weekday_yn_d0:act_weekday_yn_d8) %>%
  pivot_longer(
    cols = starts_with("act_weekday_yn_d"),
    names_to = c("day"),
    values_to = "act_weekday_yn",
    names_prefix = "act_weekday_yn_d",
  )

datinbed <- datinbed %>% left_join(datweekend_yn, by = c("ID","day"))

datinbedmean <- datinbed %>% group_by(gender.v1) %>% summarize(mean_x = mean(inbed.time_angle_x),
mean_y = mean(inbed.time_angle_y, na.rm = T),n = n()) %>% mutate(mean_angle = atan(mean_y/mean_x))

datinbedmean

```

```

# A tibble: 3 x 8
  gender.v1 mean_x mean_y      n mean_angle mean_time mean_x_uc mean_y_uc
  <chr>      <dbl> <dbl> <int>      <dbl>      <dbl>      <dbl>      <dbl>
1 female    0.858  0.117  2286      0.135      0.517      0.991      0.135
2 male      0.753  0.229  1305      0.295      1.13       0.957      0.291
3 <NA>      0.976  0.119   36      0.122      0.465      0.993      0.121

```

```

ggplot(
  aes(
    x = inbed.time_angle_x,
    y = -inbed.time_angle_y,
    color = as.factor(gender.v1)
  )
)

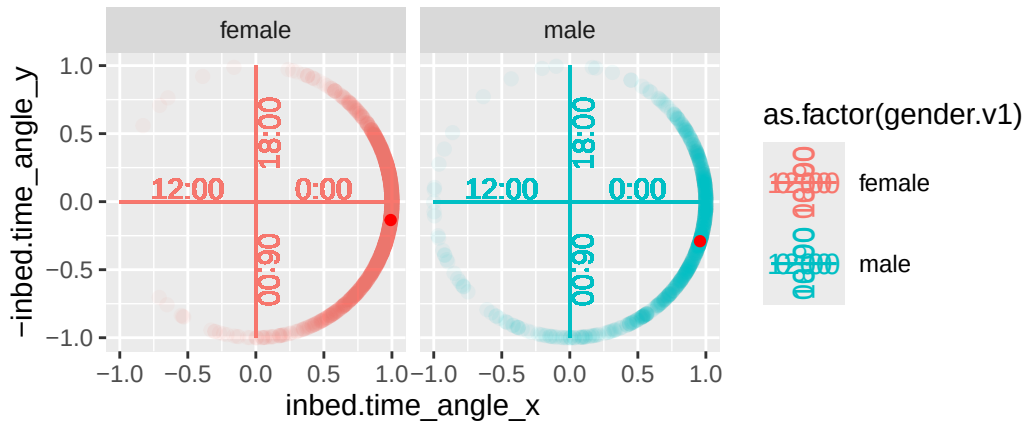
```

```

),
data = datinbed %>% filter(gender.v1 %in% c("male", "female"))
) + geom_point(alpha = 0.05, size = 2) + coord_fixed() + facet_grid( ~ gender.v1) + geom_point(
aes(x = mean_x_uc, y = -mean_y_uc),
color = "red",
data = datinbedmean %>% filter(gender.v1 %in% c("male", "female"))
) + geom_segment(x = 0,
y = 0,
xend = 1,
yend = 0) + geom_text(x = 0.5, y = 0.1, label = "0:00") +
geom_segment(x = 0,
y = 0,
xend = -1,
yend = 0) + geom_text(x = -0.5, y = 0.1, label = "12:00") +
geom_segment(x = 0,
y = 0,
xend = 0,
yend = -1) + geom_text(x = 0.1, y = -0.5, label = "06:00", angle = -90) +
geom_segment(x = 0,
y = 0,
xend = 0,
yend = 1) + geom_text(x = 0.1, y = 0.5, label = "18:00", angle = 90)

```

Warning: Removed 1147 rows containing missing values or values outside the scale range (`geom\_point()`).



```

#Pivot longer
#DVs (sleep variables with models run for each sleep DV: 8 total):

```

```

# -act_totsleep.min
# -act.outbed.time
# -act.onset.time
# -act_efficiency
# -act_awakening.min
#Weekly level:
#Variability in:
#   -act_totsleep.min_sd
#   -act.outbed.time.min_sd
#   -act.onset.time_min.sd
##test
#Pivoting longer example:
# test <- dat %>%
#   select(ID, gender.v1,
#           act_inbed.time_d0:act_inbed.time_d8,
#           act_weekday_yn_d0:act_weekday_yn_d8,
#           act_onset.time_d0: act_onset.time_d8) %>%
#   pivot_longer(
#     cols = -c(ID,gender.v1),
#     cols_vary = "slowest",
#     names_to = c(".value","day"),
#     names_pattern = "(.*)_d(.*)"
#   )
#####
#Total Sleep
#####z_ED_nat; z_HE_nat; z_SE_nat
datlong_resp <- dat %>%
  filter(act_patotaldays >= 5 & gender.v1 %in% c("male","female")) %>%
  select(
    ID,
    RecordedDate,
    pdsMF.v1,
    cg1.age.v1,
    income.v1,
    cg1_edu_grp,
    z_COI_nat,
    z_ED_nat,
    z_HE_nat,
    z_SE_nat,
    z_ED_EC_nat,
    z_ED_EL_nat,
    z_ED_ER_nat,

```

```

z_ED_SP_nat,
z_HE_EP_nat,
z_HE_HE_nat,
z_HE_HR_nat,
z_HE_SE_nat,
z_SE_EI_nat,
z_SE_EO_nat,
z_SE_ER_nat,
z_SE_HQ_nat,
z_SE_SR_nat,
z_SE_WL_nat,
act_totsleep.min_d0:act_totsleep.min_d8,
act_outbed.time_d0:act_outbed.time_d8,
act_onset.time_d0: act_onset.time_d8,
act_efficiency_d0: act_efficiency_d8,
act_weekday_yn_d0:act_weekday_yn_d8,
act_awakening.min_d0: act_awakening.min_d8) %>%
pivot_longer(
  cols = -c(ID,
             RecordedDate,
             pdsMF.v1,
             cg1.age.v1,
             income.v1,
             cg1_edu_grp,
             z_COI_nat,
             z_ED_nat,
             z_HE_nat,
             z_SE_nat,
             z_ED_EC_nat,
             z_ED_EL_nat,
             z_ED_ER_nat,
             z_ED_SP_nat,
             z_HE_EP_nat,
             z_HE_HE_nat,
             z_HE_HR_nat,
             z_HE_SE_nat,
             z_SE_EI_nat,
             z_SE_EO_nat,
             z_SE_ER_nat,
             z_SE_HQ_nat,
             z_SE_SR_nat,
             z_SE_WL_nat),

```

```

    cols_vary = "slowest",
    names_to = c(".value", "day"),
    names_pattern = "(.*)_d(.*)"
  )

datlong_weekday_yn <- dat %>%
  filter(act_patotaldays >= 5 & gender.v1 %in% c("male", "female")) %>%
  select(ID, pdsMF.v1, cg1.age.v1, income.v1, cg1.edu.deg.v1, act_weekday_yn_d0:act_weekday_yn_d1) %>%
  pivot_longer(cols = starts_with("act_weekday_yn_d"), names_prefix = "act_weekday_yn_d", names_to = c(".value", "day"))

#Weekend indicator should be Fri/Sat instead of a weekend (Sat/Sun) indicator
datlong <- datlong_resp %>% left_join(datlong_weekday_yn %>% select(ID, day, weekday_yn), by = c("ID", "day"))
datlong <- datlong %>% group_by(ID) %>% mutate(I_frisat = 1 - lead(weekday_yn))
datlong <- datlong %>% group_by(ID) %>% mutate(I_frisat = ifelse(is.na(I_frisat), lag(I_frisat), I_frisat))
#datlong$Date <- do.call(rbind, strsplit(datlong$RecordedDate, " "))[1,]
#datlong$Date <- mdy(datlong$Date)
#datlong$dayofweek <- wday(datlong$Date)

#Fix time formatting
#I cut off the day at 3pm. We could choose a different cut off
datlong <- datlong %>% ungroup() %>% mutate(
  act_outbed.time_numeric = as.numeric(do.call(rbind, strsplit(datlong$act_outbed.time, ":"))[,1]),
  act_onset.time_numeric = as.numeric(do.call(rbind, strsplit(datlong$act_onset.time, ":"))[,1])
  mutate(act_onset.time_numeric = ifelse(act_onset.time_numeric >= 17, act_onset.time_numeric - 12, act_onset.time_numeric))

datlong <- datlong %>% mutate(income.v1.10000 = income.v1 / 10000)

datlong %>% group_by(I_frisat, pdsMF.v1) %>%
  summarize(onset_mean = mean(act_onset.time_numeric, na.rm = T),
            outbed_mean = mean(act_outbed.time_numeric, na.rm = T))

```

`summarise()` has grouped output by 'I_frisat'. You can override using the `.groups` argument.

```

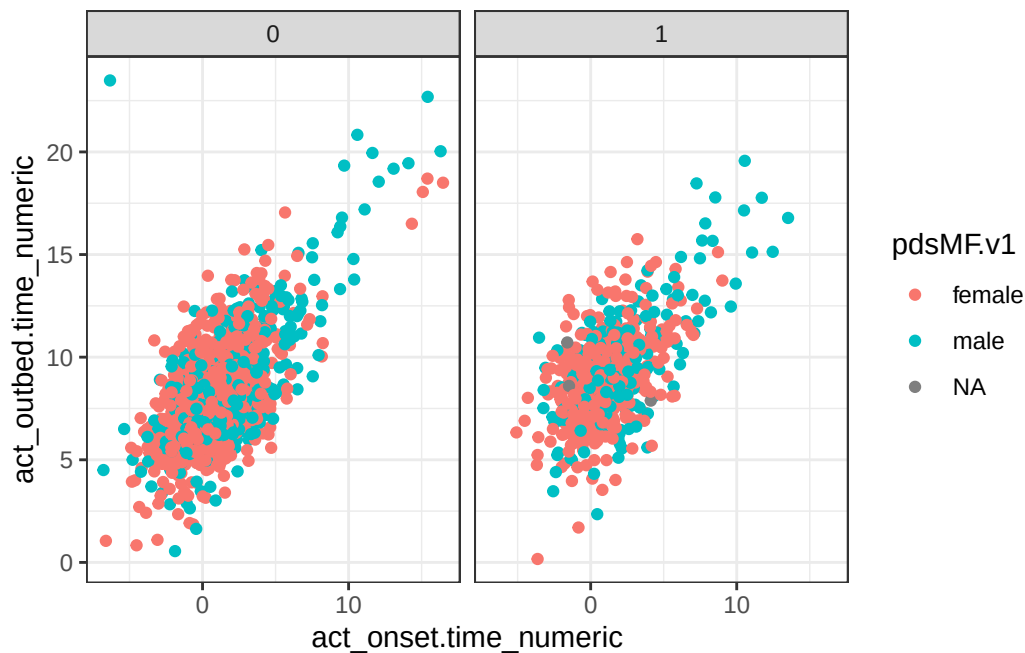
# A tibble: 6 x 4
# Groups:   I_frisat [2]
  I_frisat pdsMF.v1 onset_mean outbed_mean
    <dbl> <chr>         <dbl>         <dbl>
1         0 female      0.487          7.79

```

2	0 male	1.30	8.18
3	0 <NA>	-0.289	6.64
4	1 female	0.990	8.93
5	1 male	2.07	9.68
6	1 <NA>	0.204	8.85

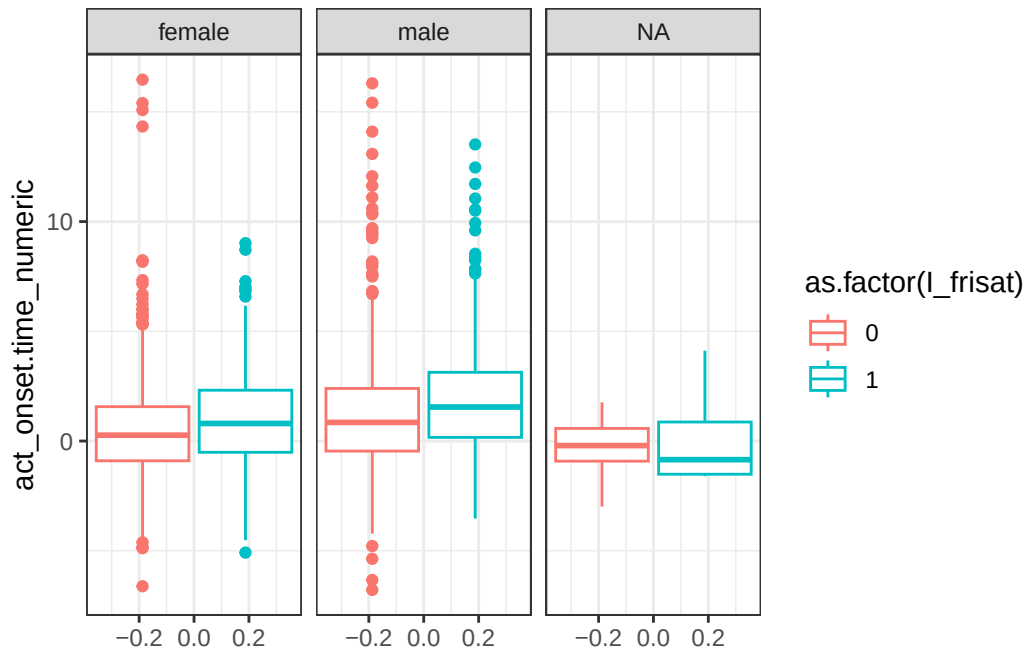
```
ggplot(aes(x = act_onset.time_numeric,
           y = act_outbed.time_numeric,
           color = pdsMF.v1), data = datlong) + geom_point() + theme_bw() + facet_grid(~I_fr
```

Warning: Removed 569 rows containing missing values or values outside the scale range (`geom\_point()`).



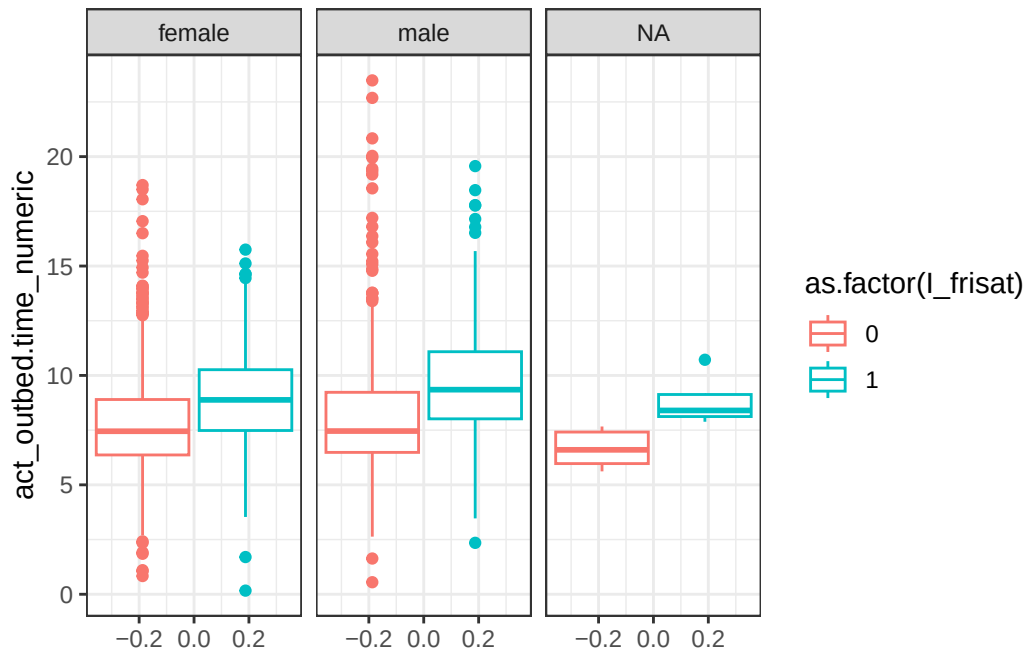
```
ggplot(aes(y = act_onset.time_numeric,
           color = as.factor(I_frisat)), data = datlong) + geom_boxplot() + theme_bw() + fac
```

Warning: Removed 569 rows containing non-finite outside the scale range (`stat\_boxplot()`).



```
ggplot(aes(y = act_outbed.time_numeric,
           color = as.factor(I_frisat)), data = datlong) + geom_boxplot() + theme_bw() + fac
```

Warning: Removed 569 rows containing non-finite outside the scale range (`stat\_boxplot()`).

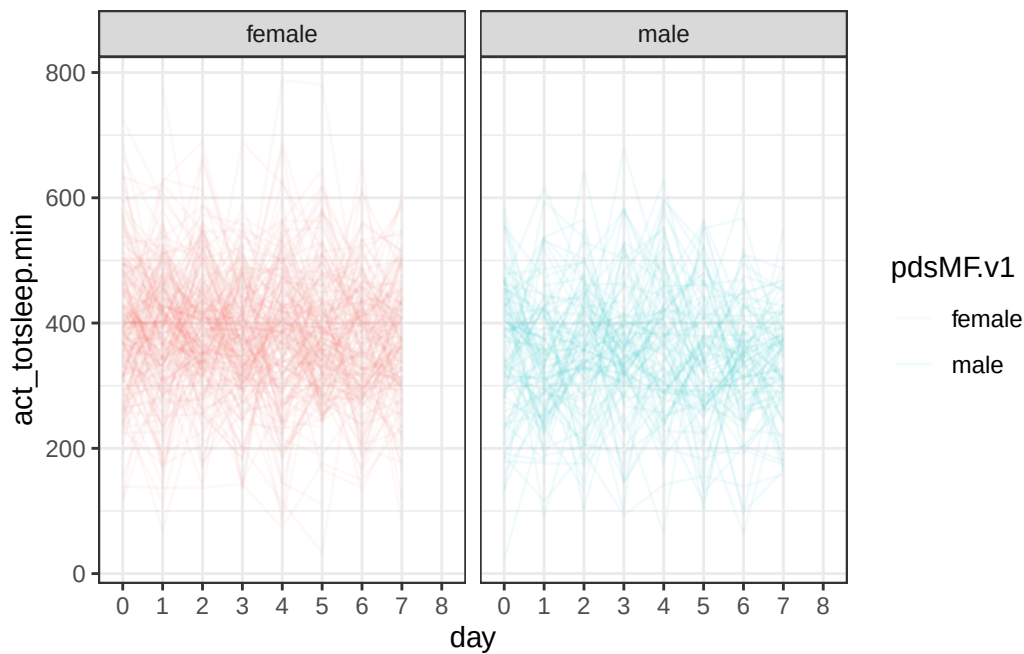


Models

Total Sleep

```
ggplot(aes(x = day, y = act_totsleep.min, group = ID, color = pdsMF.v1), data = datlong %>%
```

Warning: Removed 450 rows containing missing values or values outside the scale range (``geom_line()``).



```
library(lme4)
#####
#act_totalsleep.min
#####
#####
#Model 1: COI Total
#####
mod1_totsleep <- lmer(act_totsleep.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_g
summary(mod1_totsleep)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [


```
lmerModLmerTest]
```

```
Formula: act_totsleep.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +  
        cg1_edu_grp + I_frisat + z_COI_nat + (1 | ID)
```

```
Data: datlong
```

```
REML criterion at convergence: 25021
```

```
Scaled residuals:
```

Min	1Q	Median	3Q	Max
-3.4209	-0.5975	-0.0158	0.5723	3.5360

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2393	48.92
Residual		7878	88.76

Number of obs: 2094, groups: ID, 291

```
Fixed effects:
```

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	411.7789	18.7976	284.9492	21.906	< 2e-16 ***
pdsMF.v1male	-28.6954	7.4015	280.9184	-3.877	0.000132 ***
cg1.age.v1	-0.7625	0.3715	282.1889	-2.052	0.041075 *
income.v1.10000	0.2520	1.2129	277.0686	0.208	0.835580
cg1_edu_grp<=BS	-8.0561	8.2250	287.3229	-0.979	0.328176
cg1_edu_grp>=MS	-6.3878	14.0958	283.3905	-0.453	0.650770
I_frisat	33.3106	4.3734	1830.0143	7.617	4.15e-14 ***
z_COI_nat	0.3572	4.9128	286.2299	0.073	0.942083

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
```

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst
pdsMF.v1mal	-0.124						
cg1.age.v1	-0.884	0.014					
inc.1.10000	-0.203	0.011	-0.022				
cg1_d_g<=BS	-0.267	-0.067	-0.002	-0.163			
cg1_d_g>=MS	-0.071	-0.140	-0.049	-0.304	0.471		
I_frisat	-0.060	0.005	-0.004	0.008	-0.005	-0.015	
z_COI_nat	0.217	-0.014	-0.040	-0.140	-0.078	-0.115	0.009

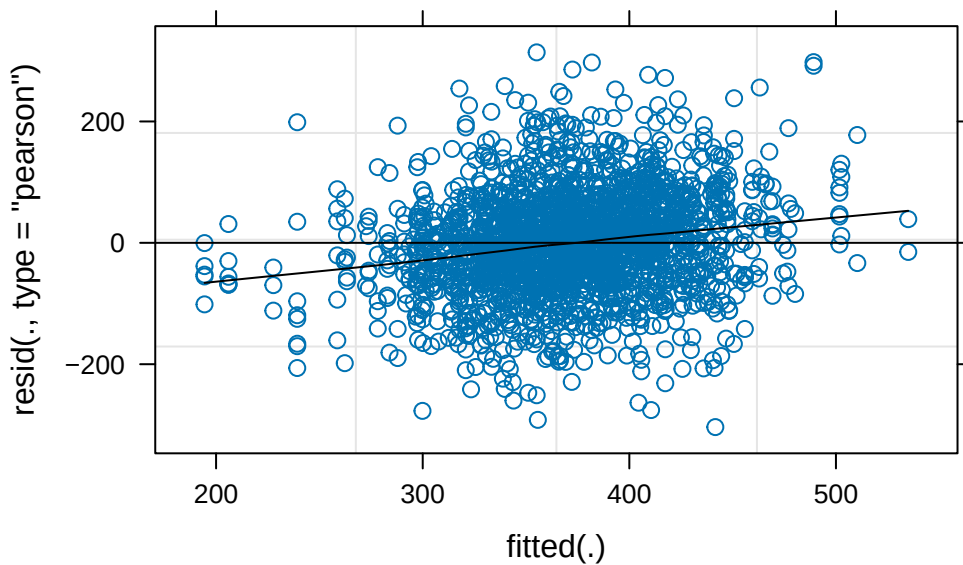
```
ci <- confint(mod1_totsleep)
```

Computing profile confidence intervals ...

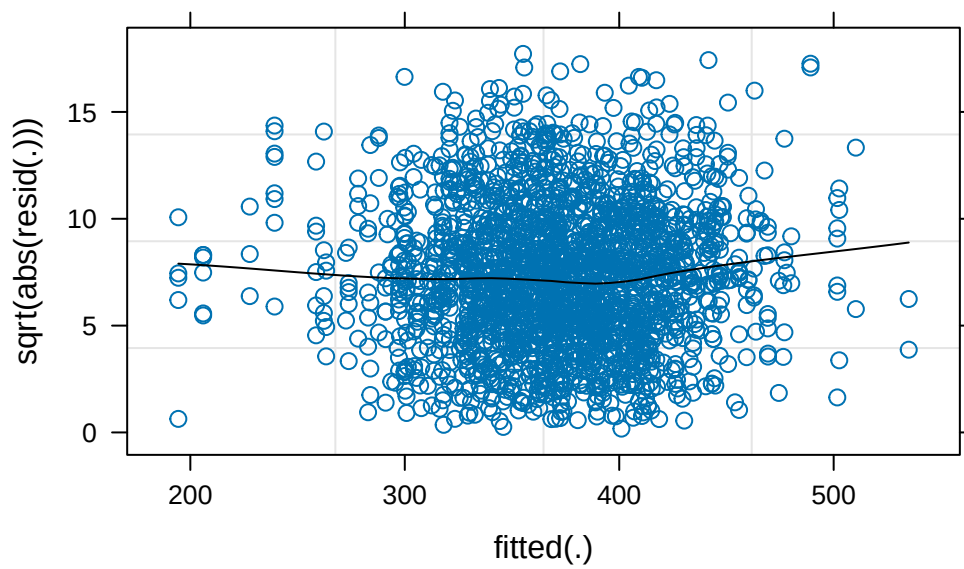
```
print(ci)
```

	2.5 %	97.5 %
.sig01	42.455328	54.18632217
.sigma	85.913916	91.70655844
(Intercept)	375.270755	448.30241053
pdsMF.v1male	-43.071569	-14.31672341
cg1.age.v1	-1.484586	-0.04109889
income.v1.10000	-2.104038	2.60802287
cg1_edu_grp<=BS	-24.025910	7.93133648
cg1_edu_grp>=MS	-33.762555	21.00121401
I_frisat	24.735738	41.88188549
z_COI_nat	-9.192216	9.89576657

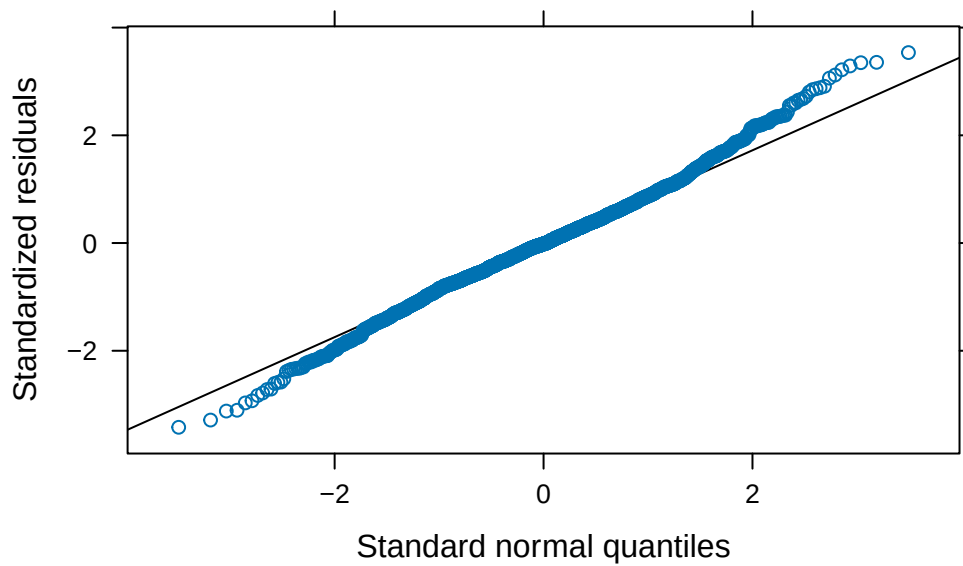
```
plot(mod1_totsleep, type=c("p", "smooth"), col.line=1)
```



```
plot(mod1_totsleep,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p", "smooth"), col.line=1)
```



```
lattice::qqmath(mod1_totsleep)
```



```
#Table
#Fixed effects
mod1_totsleep_table <- data.frame(
  predictor = attr(summary(mod1_totsleep)$coef,"dimnames")[[1]],
  B = summary(mod1_totsleep)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod1_totsleep)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_totsleep)$CI_low,4),", ",r
pval = summary(mod1_totsleep)$coef[,5]
)

mod1_totsleep_table[,c("B","beta","pval")] <-
round(mod1_totsleep_table[,c("B","beta","pval")],4)
write.csv(mod1_totsleep_table,file = "../bohnert_sleep_git/mod1_totsleep_table.csv")

#Random effects
#within
sigma2 <- summary(mod1_totsleep)$sigma^2
#between
tau00 <- attr(summary(mod1_totsleep)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_totsleep)$devcomp$dims["q"]
obs <- summary(mod1_totsleep)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_totsleep)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_totsleep)[,"R2c"]
mod1_totsleep_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_totsleep_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	7877.64	2393.244	0.2330125	291	2094	0.04440931	0.2670739

```

#####
#Model 2: COI Sub-domains
#####
mod2_totsleep <- lmer(act_totsleep.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_g
summary(mod2_totsleep)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: act_totsleep.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)

Data: datlong

REML criterion at convergence: 25008

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.4164	-0.5979	-0.0184	0.5715	3.5309

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2406	49.05
	Residual	7877	88.75

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	408.4329	19.3787	282.2139	21.076	< 2e-16 ***
pdsMF.v1male	-28.5163	7.4326	278.7177	-3.837	0.000154 ***
cg1.age.v1	-0.7515	0.3730	280.4664	-2.015	0.044867 *
income.v1.10000	0.4730	1.2317	276.2241	0.384	0.701274
cg1_edu_grp<=BS	-7.6850	8.2546	285.5751	-0.931	0.352643
cg1_edu_grp>=MS	-5.9441	14.1277	281.5098	-0.421	0.674265
I_frisat	33.3302	4.3733	1830.0485	7.621	4e-14 ***
z_ED_nat	7.8940	8.9202	288.7548	0.885	0.376915
z_HE_nat	3.3644	8.4003	277.9124	0.401	0.689087
z_SE_nat	-7.4929	7.4189	283.2051	-1.010	0.313370

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

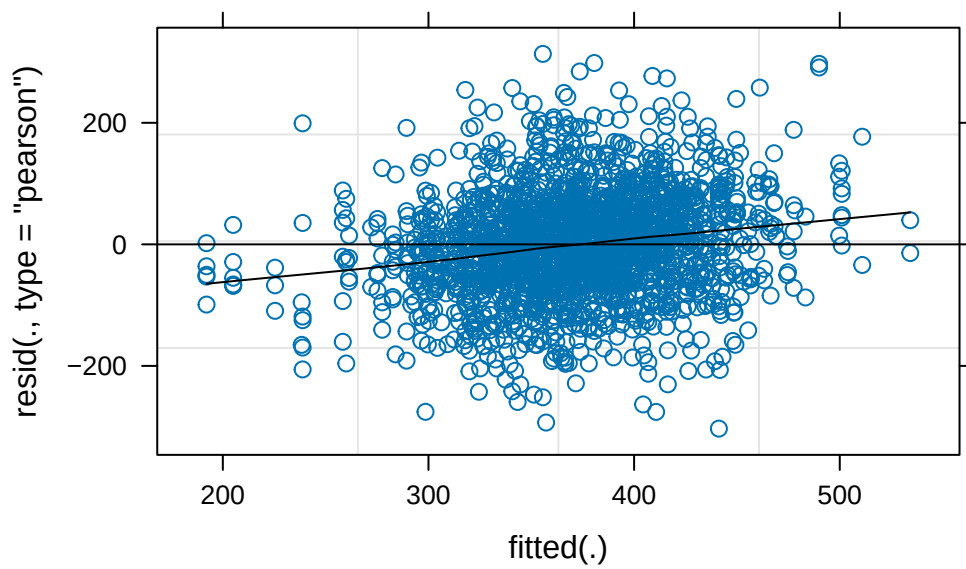
Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
pdsMF.v1male	-0.112								
cg1.age.v1	-0.872	0.010							
inc.1.10000	-0.211	0.017	-0.021						
cg1_d_g<=BS	-0.257	-0.063	-0.003	-0.152					
cg1_d_g>=MS	-0.076	-0.140	-0.047	-0.296	0.471				
I_frisat	-0.059	0.005	-0.003	0.009	-0.005	-0.015			
z_ED_nat	0.019	0.054	-0.021	0.130	0.045	-0.009	0.003		
z_HE_nat	-0.202	-0.055	0.061	0.005	-0.035	0.017	0.005	-0.241	
z_SE_nat	0.209	-0.029	-0.035	-0.200	-0.067	-0.065	-0.001	-0.712	-0.256

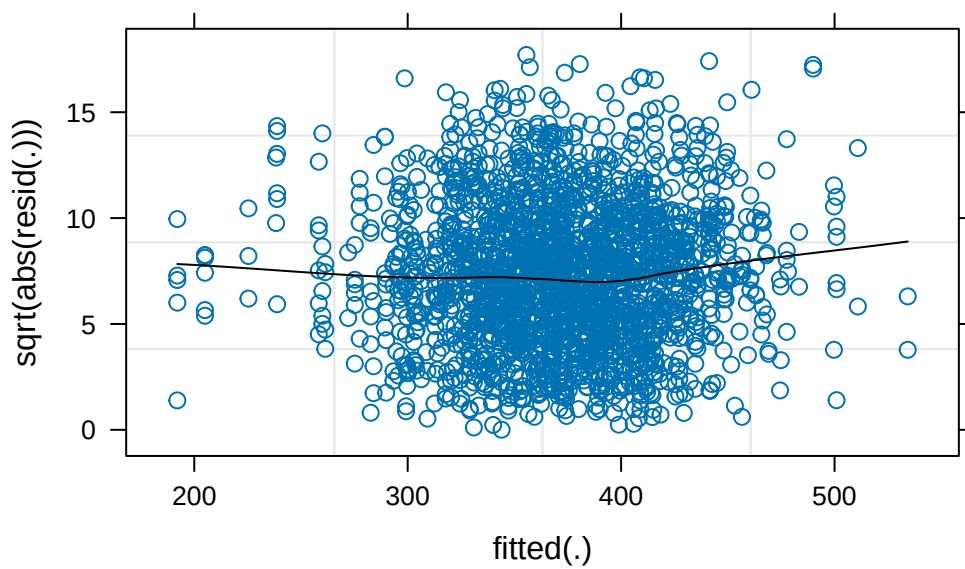
```
ci <- confint(mod2_totsleep)
```

Computing profile confidence intervals ...

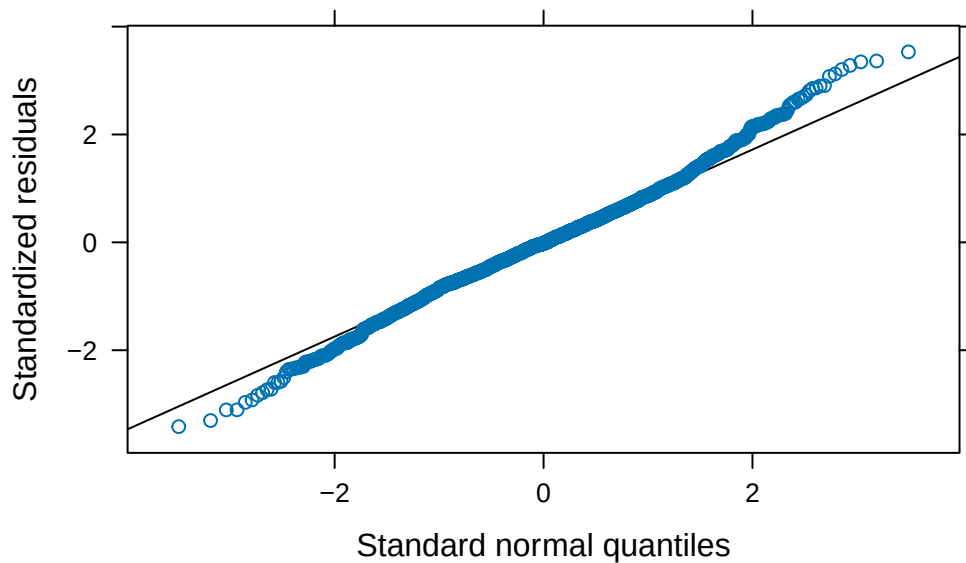
```
plot(mod2_totsleep, type=c("p","smooth"), col.line=1)
```



```
plot(mod2_totsleep,
     sqrt(abs(resid(.)))~fitted(.),
     type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_totsleep)
```



```

mod2_totsleep_table <- data.frame(
  predictor = attr(summary(mod2_totsleep)$coef,"dimnames")[[1]],
  B = summary(mod2_totsleep)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod2_totsleep)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_totsleep)$CI_low,4),", ",round(effectsize::standardize_parameters(mod2_totsleep)$CI_high,4),")"),
  pval = summary(mod2_totsleep)$coef[,5]
)

mod2_totsleep_table[,c("B","beta","pval")] <-
round(mod2_totsleep_table[,c("B","beta","pval")],4)
write.csv(mod2_totsleep_table,file = "../bohnert_sleep_git/mod2_totsleep_table.csv")

#Random effects
#within
sigma2 <- summary(mod2_totsleep)$sigma^2
#between
tau00 <- attr(summary(mod2_totsleep)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_totsleep)$devcomp$dims["q"]
obs <- summary(mod2_totsleep)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_totsleep)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_totsleep)[,"R2c"]
mod2_totsleep_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_totsleep_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	7876.802	2405.565	0.2339505	291	2094	0.04556391	0.2688547

```
#####
#Model 3: COI Sub-domains
#####
mod3_totsleep <- lmer(act_totsleep.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_g
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + + (1|ID) , data = datlong)
summary(mod3_totsleep)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: act_totsleep.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +
cg1_edu_grp + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat +
+(1 | ID)
Data: datlong

REML criterion at convergence: 21719.2

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-3.1955	-0.6080	-0.0505	0.5773	3.5406

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2112	45.95
Residual		8259	90.88

Number of obs: 1821, groups: ID, 255

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	418.7380	21.5858	231.4723	19.399	< 2e-16 ***
pdsMF.v1male	-32.2878	7.8324	230.9633	-4.122	5.23e-05 ***
cg1.age.v1	-0.8726	0.3818	232.3485	-2.286	0.0232 *
income.v1.10000	-0.1109	1.3030	228.5421	-0.085	0.9323
cg1_edu_grp<=BS	-13.8137	8.8150	235.8752	-1.567	0.1184
cg1_edu_grp>=MS	-8.0572	14.8504	233.6772	-0.543	0.5880
z_ED_EC_nat	3.3091	4.3403	228.4369	0.762	0.4466
z_ED_EL_nat	6.2462	9.8483	236.7664	0.634	0.5265
z_ED_ER_nat	-12.3019	17.0084	234.0396	-0.723	0.4702
z_ED_SP_nat	-0.4341	9.3049	234.0697	-0.047	0.9628
z_HE_EP_nat	89.4098	37.9568	234.2728	2.356	0.0193 *
z_HE_HE_nat	21.7552	21.5020	235.1092	1.012	0.3127
z_HE_HR_nat	-9.2291	8.8142	231.6098	-1.047	0.2962
z_HE_SE_nat	8.0662	6.1144	236.5626	1.319	0.1884
z_SE_EI_nat	4.0064	16.8845	234.3643	0.237	0.8126
z_SE_EO_nat	-8.1434	11.2727	235.3090	-0.722	0.4708
z_SE_ER_nat	-2.4398	11.5365	234.1299	-0.211	0.8327
z_SE_HQ_nat	-8.3699	10.6252	231.8531	-0.788	0.4317
z_SE_SR_nat	-10.3993	10.1205	232.0055	-1.028	0.3052
z_SE_WL_nat	19.5604	12.1617	232.4145	1.608	0.1091

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

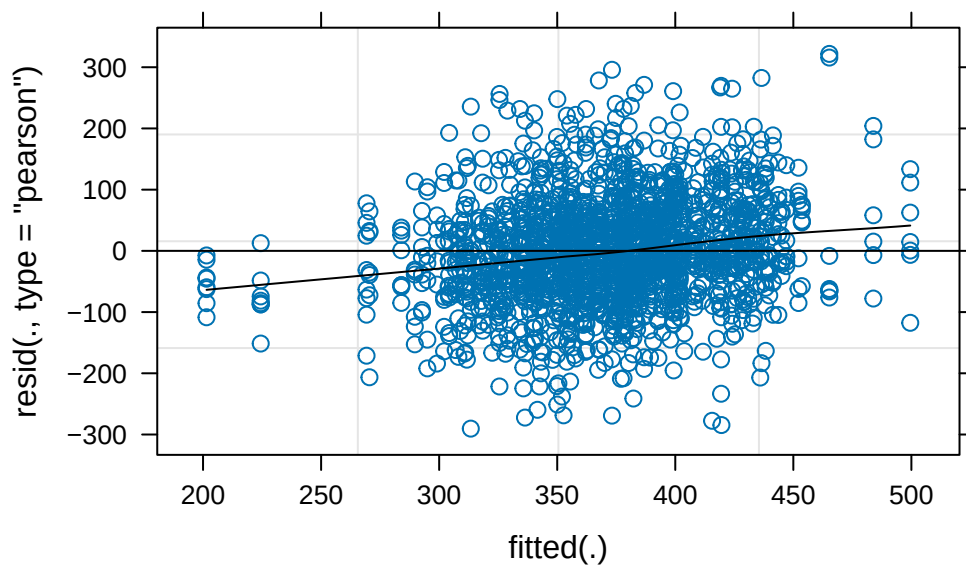
Correlation matrix not shown by default, as $p = 20 > 12$.

Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

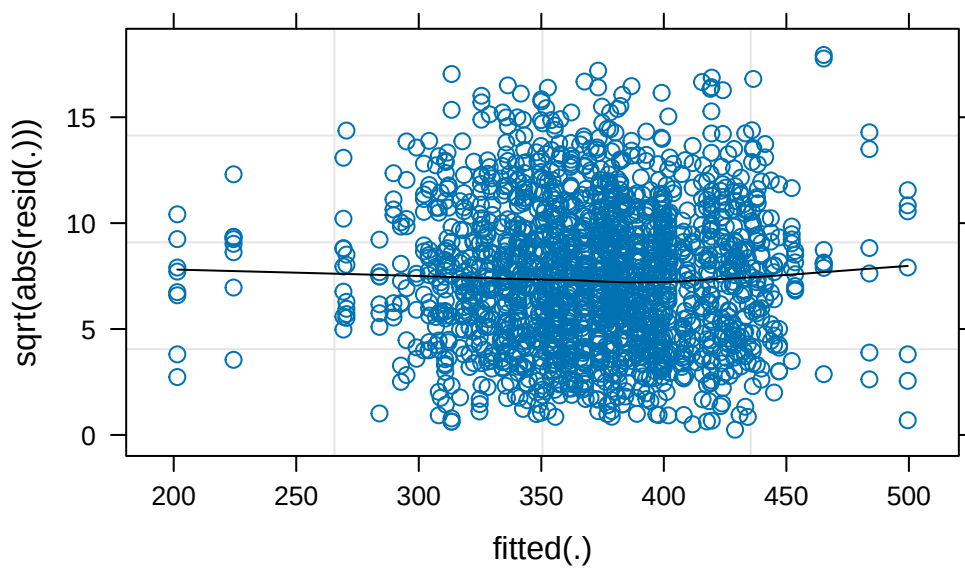
```
ci <- confint(mod3_totsleep)
```

Computing profile confidence intervals ...

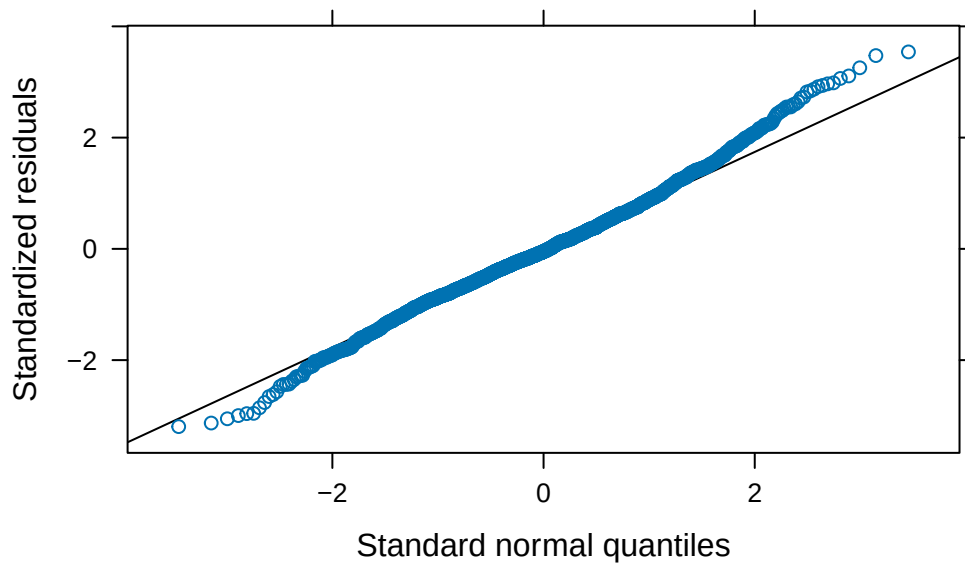
```
plot(mod3_totsleep, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_totsleep,
     sqrt(abs(resid(.)))~fitted(.),
     type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_totsleep)
```



```
#Table
#Fixed Effects
mod3_totsleep_table <- data.frame(
  predictor = attr(summary(mod3_totsleep)$coef,"dimnames")[[1]],
  B = summary(mod3_totsleep)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod3_totsleep)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_totsleep)$CI_low,4),", ",round(effectsize::standardize_parameters(mod3_totsleep)$CI_high,4),")"),
  pval = summary(mod3_totsleep)$coef[,5]
)

mod3_totsleep_table[,c("B","beta","pval")] <-
round(mod3_totsleep_table[,c("B","beta","pval")],4)
write.csv(mod3_totsleep_table,file = "../bohnert_sleep_git/mod3_totsleep_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_totsleep)$sigma^2
#between
tau00 <- attr(summary(mod3_totsleep)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_totsleep)$devcomp$dims["q"]
obs <- summary(mod3_totsleep)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_totsleep)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_totsleep)[,"R2c"]
mod3_totsleep_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
```

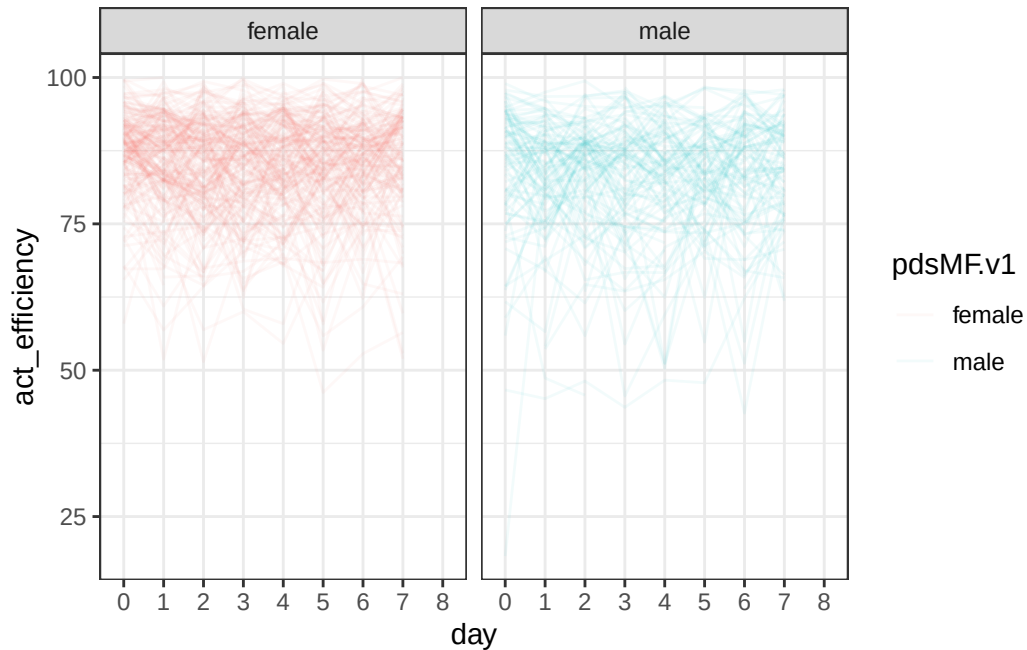
```
mod3_totsleep_table_re
```

```
          sigma2    tau00      ICC    N obs      R2marg    R2cond  
(Intercept) 8258.58 2111.626 0.2036243 255 1821 0.04720748 0.2412192
```

Sleep Efficiency

```
ggplot(aes(x = day, y = act_efficiency, group = ID, color = pdsMF.v1), data = datlong %>% fi
```

Warning: Removed 450 rows containing missing values or values outside the scale range (`geom\_line()`).



```
library(lme4)  
#####  
#act_efficiency  
#####  
#####  
#Model 1: COI Total  
#####  
mod1_eff <- lmer(act_efficiency ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp + I_  
summary(mod1_eff)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula:

act_efficiency ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp +
I_frisat + z_COI_nat + (1 | ID)

Data: datlong

REML criterion at convergence: 14497.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-8.2034	-0.4676	0.1227	0.5771	2.8974

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	39.39	6.277
	Residual	45.44	6.741

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	89.46227	2.14382	286.07361	41.730	< 2e-16 ***
pdsMF.v1male	-2.05637	0.84636	282.81463	-2.430	0.01574 *
cg1.age.v1	-0.12700	0.04245	283.64255	-2.992	0.00302 **
income.v1.10000	0.33833	0.13901	280.51860	2.434	0.01556 *
cg1_edu_grp<=BS	0.01410	0.93669	287.66925	0.015	0.98800
cg1_edu_grp>=MS	-1.00832	1.60956	284.30556	-0.626	0.53152
I_frisat	0.46711	0.33287	1816.16871	1.403	0.16070
z_COI_nat	-0.27012	0.56012	285.87692	-0.482	0.63000

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst
pdsMF.v1mal	-0.124						
cg1.age.v1	-0.886	0.015					
inc.1.10000	-0.206	0.010	-0.020				
cg1_d_g<=BS	-0.263	-0.070	-0.002	-0.165			
cg1_d_g>=MS	-0.065	-0.142	-0.053	-0.305	0.468		
I_frisat	-0.040	0.003	-0.003	0.006	-0.004	-0.011	
z_COI_nat	0.215	-0.013	-0.039	-0.142	-0.072	-0.108	0.006

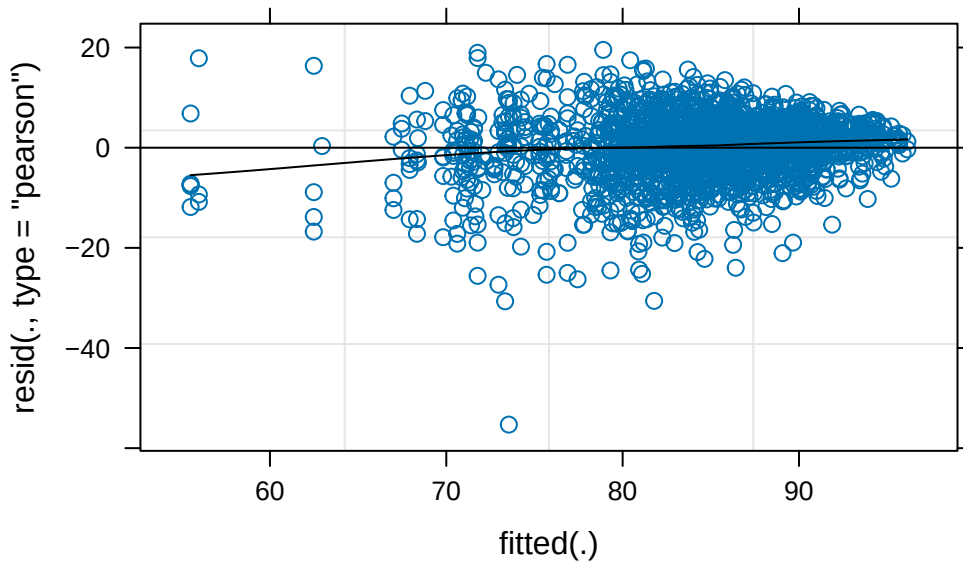
```
ci <- confint(mod1_eff)
```

Computing profile confidence intervals ...

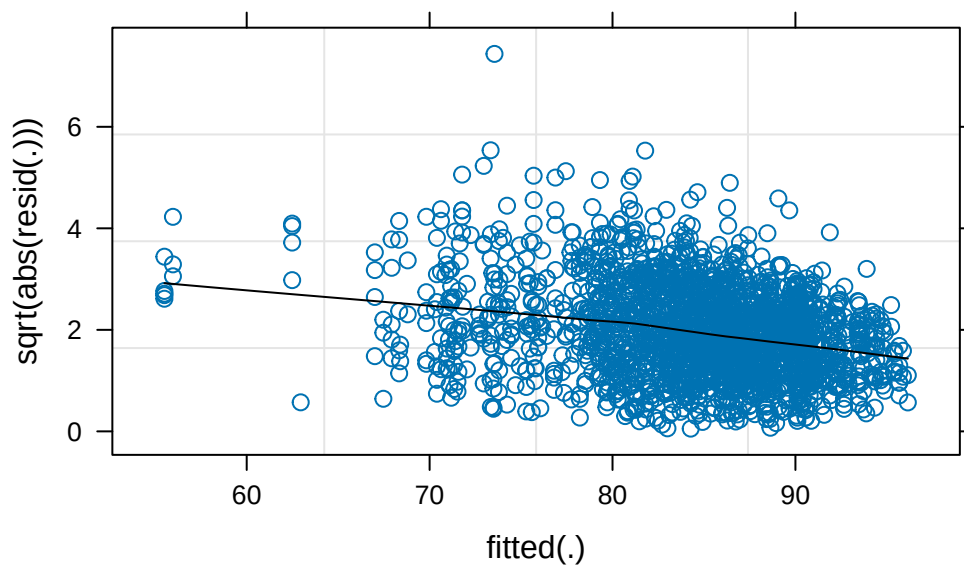
```
ci
```

	2.5 %	97.5 %
.sig01	5.6319141	6.81564071
.sigma	6.5248193	6.96487497
(Intercept)	85.2962974	93.62585899
pdsMF.v1male	-3.7006222	-0.41232707
cg1.age.v1	-0.2094534	-0.04450856
income.v1.10000	0.0683089	0.60837397
cg1_edu_grp<=BS	-1.8053552	1.83405708
cg1_edu_grp>=MS	-4.1349296	2.11864674
I_frisat	-0.1860866	1.11902051
z_COI_nat	-1.3581397	0.81811588

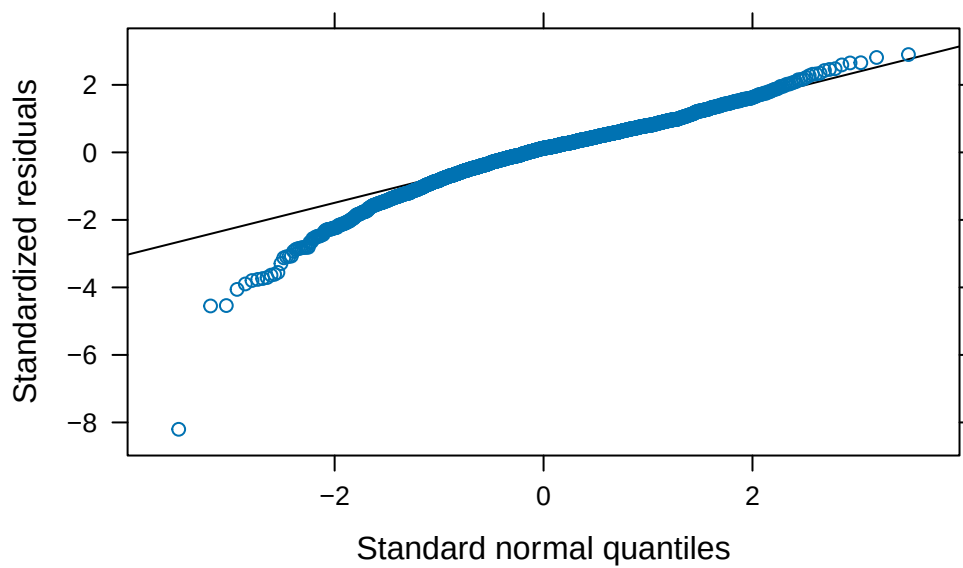
```
plot(mod1_eff, type=c("p","smooth"), col.line=1)
```



```
plot(mod1_eff,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod1_eff)
```



```
#Table
#fixed effects
mod1_eff_table <- data.frame(
  predictor = attr(summary(mod1_eff)$coef,"dimnames")[[1]],
  B = summary(mod1_eff)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod1_eff)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_eff)$CI_low,4),", ",round(
pval = summary(mod1_eff)$coef[,5]
)

mod1_eff_table[,c("B","beta","pval")] <-
round(mod1_eff_table[,c("B","beta","pval")],4)
write.csv(mod1_eff_table,file = "../bohnert_sleep_git/mod1_eff_table.csv")

#Random effects
#within
sigma2 <- summary(mod1_eff)$sigma^2
#between
tau00 <- attr(summary(mod1_eff)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_eff)$devcomp$dims["q"]
obs <- summary(mod1_eff)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_eff)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_eff)[,"R2c"]
mod1_eff_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_eff_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	45.43766	39.39478	0.4643834	291	2094	0.03802443	0.4847499

```

#####
#Model 2: COI Sub-domains
#####
mod2_eff <- lmer(act_efficiency ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp + I.
summary(mod2_eff)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula:

```
act_efficiency ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp +
  I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)
```

Data: datlong

REML criterion at convergence: 14493.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-8.2014 -0.4700 0.1207 0.5795 2.8912

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	39.49	6.284
Residual		45.44	6.741

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	88.85867	2.20970	283.78449	40.213	< 2e-16 ***
pdsMF.v1male	-2.11539	0.84956	280.71257	-2.490	0.01335 *
cg1.age.v1	-0.12367	0.04259	281.84080	-2.904	0.00398 **
income.v1.10000	0.34133	0.14100	279.20601	2.421	0.01612 *
cg1_edu_grp<=BS	-0.01303	0.93944	285.74371	-0.014	0.98894
cg1_edu_grp>=MS	-0.94554	1.61223	282.38311	-0.586	0.55802
I_frisat	0.46815	0.33286	1816.24237	1.406	0.15976
z_ED_nat	-0.41825	1.01358	287.20083	-0.413	0.68018
z_HE_nat	1.09553	0.95997	281.72164	1.141	0.25475
z_SE_nat	-0.43765	0.84568	283.73600	-0.518	0.60520

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
pdsMF.v1mal	-0.111								
cg1.age.v1	-0.874	0.012							
inc.1.10000	-0.214	0.016	-0.017						
cg1_d_g<=BS	-0.254	-0.066	-0.003	-0.154					
cg1_d_g>=MS	-0.071	-0.142	-0.051	-0.297	0.467				
I_frisat	-0.039	0.003	-0.002	0.006	-0.004	-0.011			
z_ED_nat	0.014	0.056	-0.016	0.127	0.047	-0.008	0.002		
z_HE_nat	-0.203	-0.059	0.060	0.004	-0.032	0.019	0.003	-0.241	
z_SE_nat	0.213	-0.028	-0.039	-0.198	-0.067	-0.064	0.000	-0.711	-0.258

```
ci <- confint(mod2_eff)
```

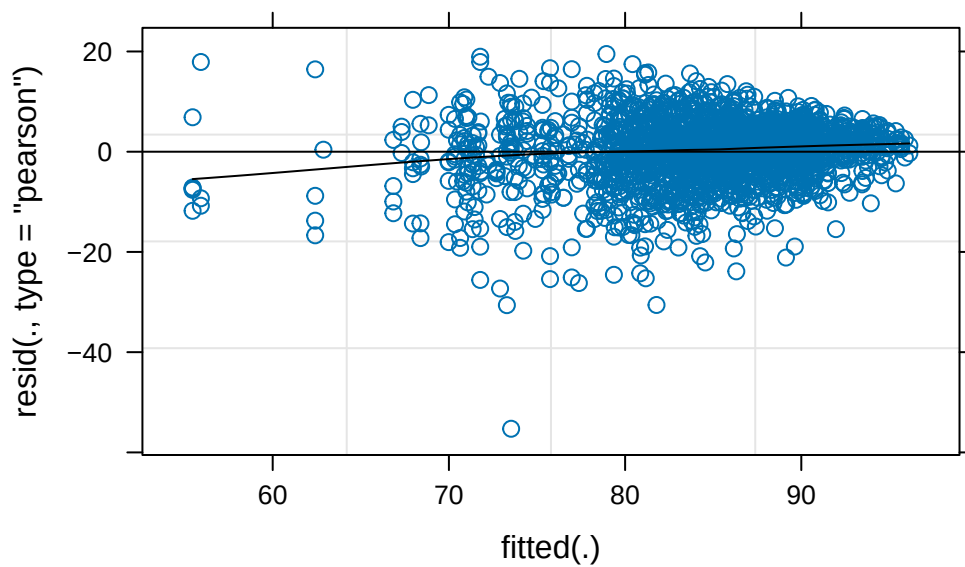
Computing profile confidence intervals ...

```
ci
```

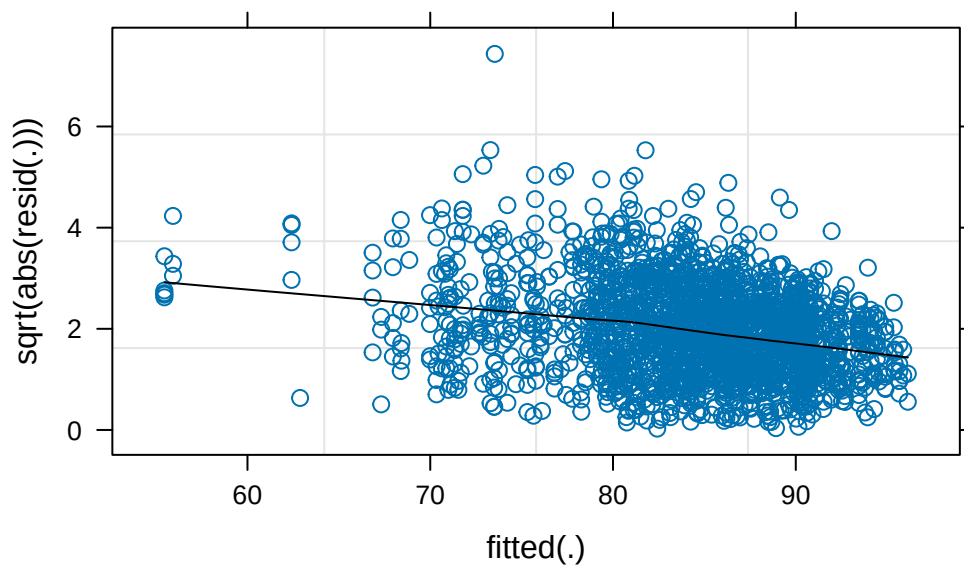
2.5 % 97.5 %

```
.sig01          5.61545975  6.79643013
.sigma          6.52468071  6.96470565
(Intercept)    84.57896667 93.13440703
pdsMF.v1male   -3.75997129 -0.47087214
cg1.age.v1     -0.20607916 -0.04119218
income.v1.10000 0.06846093  0.61431857
cg1_edu_grp<=BS -1.83126600  1.80612365
cg1_edu_grp>=MS -4.06626528  2.17571420
I_frisat       -0.18516476  1.11990658
z_ED_nat       -2.37907786  1.54554312
z_HE_nat       -0.76326151  2.95335971
z_SE_nat       -2.07555934  1.19872377
```

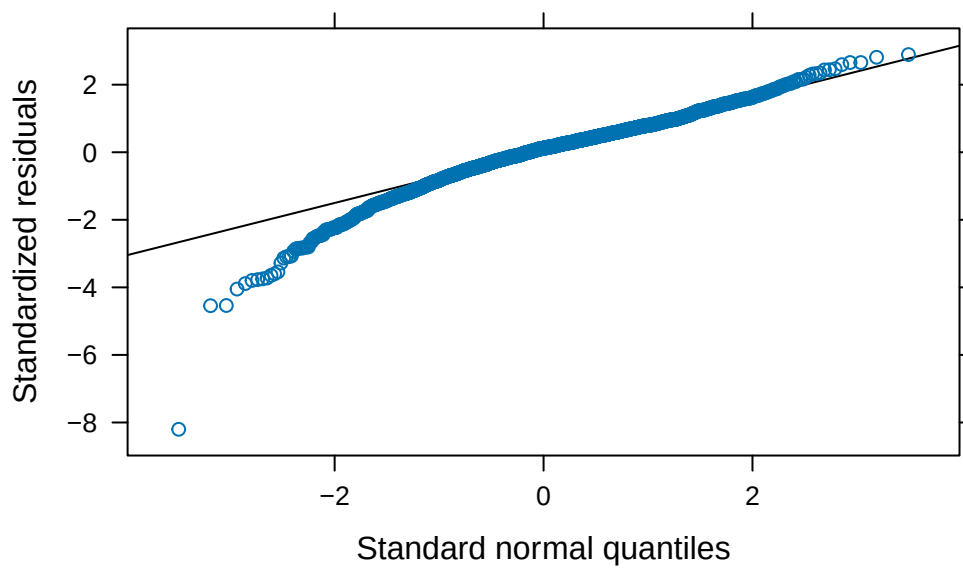
```
plot(mod2_eff, type=c("p","smooth"), col.line=1)
```



```
plot(mod2_eff,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_eff)
```



```
#Table
#fixed effects
mod2_eff_table <- data.frame(
  predictor = attr(summary(mod2_eff)$coef,"dimnames")[[1]],
  B = summary(mod2_eff)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod2_eff)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_eff)$CI_low,4),", ",round(
pval = summary(mod2_eff)$coef[,5]
)

mod2_eff_table[,c("B","beta","pval")] <-
round(mod2_eff_table[,c("B","beta","pval")],4)
write.csv(mod2_eff_table,file = "../bohnert_sleep_git/mod2_eff_table.csv")

#Random effects
#within
sigma2 <- summary(mod2_eff)$sigma^2
#between
tau00 <- attr(summary(mod2_eff)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_eff)$devcomp$dims["q"]
obs <- summary(mod2_eff)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_eff)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_eff)[,"R2c"]
mod2_eff_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_eff_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	45.43599	39.4925	0.4650089	291	2094	0.04071416	0.4867906

```

#####
#Model 3: COI Sub-domains
#####
mod3_eff <- lmer(act_efficiency ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp + z_
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_eff)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula:

act_efficiency ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp +
z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat + z_HE_EP_nat +
z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat + z_SE_EO_nat +
z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat + (1 | ID)

Data: datlong

REML criterion at convergence: 12606

Scaled residuals:

Min	1Q	Median	3Q	Max
-8.1581	-0.4713	0.1146	0.5746	2.7932

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	38.40	6.197
	Residual	46.72	6.835

Number of obs: 1821, groups: ID, 255

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	87.32082	2.52617	235.70363	34.566	< 2e-16 ***
pdsMF.v1male	-2.10789	0.91756	234.25498	-2.297	0.02249 *
cg1.age.v1	-0.12099	0.04467	235.25578	-2.708	0.00726 **
income.v1.10000	0.39506	0.15294	232.76260	2.583	0.01040 *
cg1_edu_grp<=BS	-0.90397	1.02828	238.20712	-0.879	0.38023
cg1_edu_grp>=MS	-1.85650	1.73616	235.88025	-1.069	0.28602
z_ED_EC_nat	0.03379	0.50917	233.68666	0.066	0.94715
z_ED_EL_nat	-1.30209	1.14854	237.99901	-1.134	0.25806
z_ED_ER_nat	-0.77234	1.98701	236.84957	-0.389	0.69785
z_ED_SP_nat	0.30246	1.08724	236.54340	0.278	0.78111
z_HE_EP_nat	9.59646	4.43900	235.05472	2.162	0.03164 *
z_HE_HE_nat	1.91621	2.51047	237.16155	0.763	0.44605
z_HE_HR_nat	-1.12541	1.03202	234.72484	-1.090	0.27661
z_HE_SE_nat	0.62709	0.71280	238.84958	0.880	0.37988
z_SE_EI_nat	2.73947	1.97220	236.90863	1.389	0.16612
z_SE_EO_nat	-2.45972	1.31621	236.96016	-1.869	0.06289 .
z_SE_ER_nat	-0.63665	1.34781	236.72496	-0.472	0.63711
z_SE_HQ_nat	2.50796	1.24353	235.27501	2.017	0.04485 *
z_SE_SR_nat	-1.08541	1.18450	235.15629	-0.916	0.36042
z_SE_WL_nat	0.28274	1.42320	235.09838	0.199	0.84269

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation matrix not shown by default, as $p = 20 > 12$.

Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

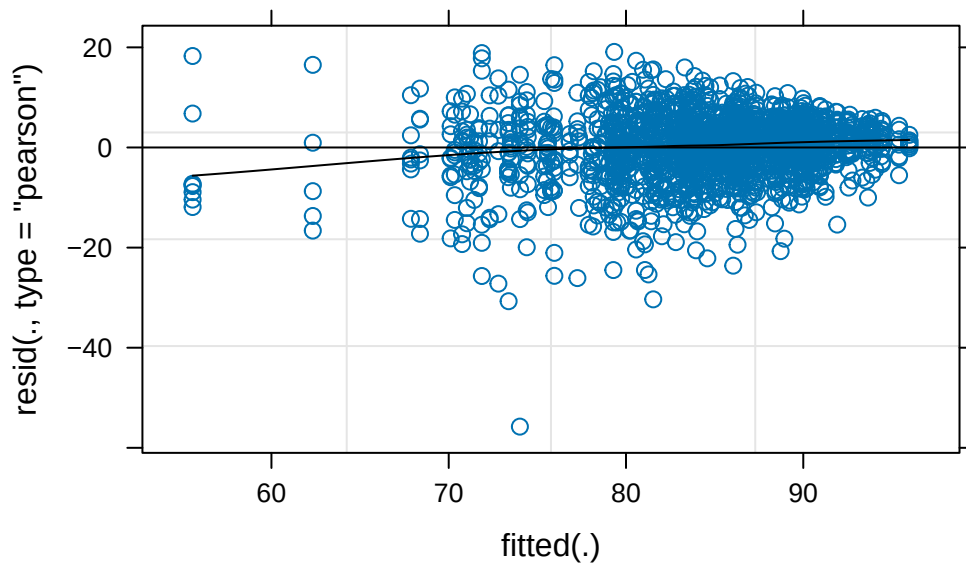
```
ci <- confint(mod3_eff)
```

Computing profile confidence intervals ...

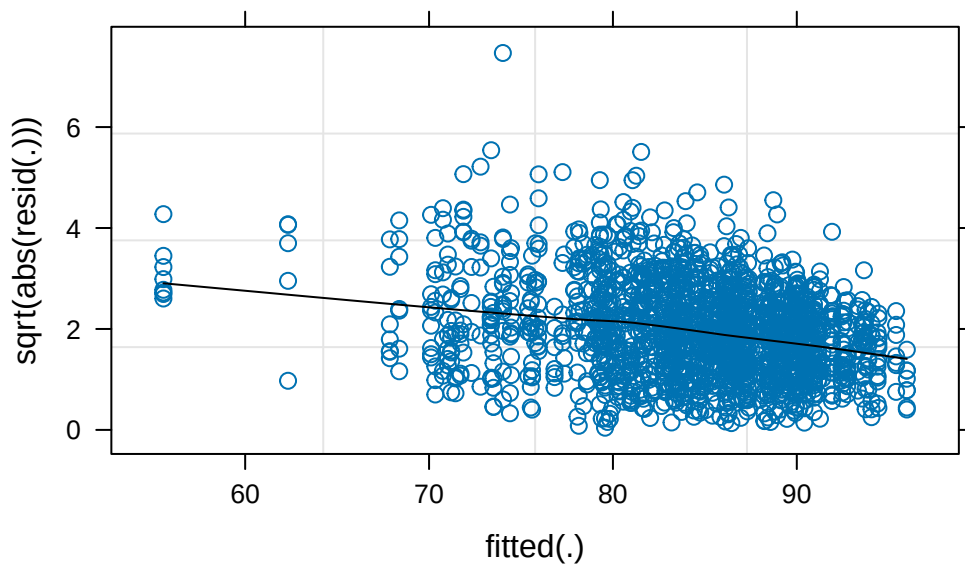
```
ci
```

	2.5 %	97.5 %
.sig01	5.3282778	6.55914042
.sigma	6.6027981	7.08164902
(Intercept)	82.5413493	92.08480901
pdsMF.v1male	-3.8411960	-0.37520701
cg1.age.v1	-0.2052061	-0.03644008
income.v1.10000	0.1064765	0.68411775
cg1_edu_grp<=BS	-2.8435800	1.04179619
cg1_edu_grp>=MS	-5.1349417	1.42401680
z_ED_EC_nat	-0.9277406	0.99553288
z_ED_EL_nat	-3.4706168	0.86904507
z_ED_ER_nat	-4.5258007	2.98135893
z_ED_SP_nat	-1.7463569	2.36136218
z_HE_EP_nat	1.2011458	17.97020356
z_HE_HE_nat	-2.8253054	6.65976812
z_HE_HR_nat	-3.0752565	0.82324045
z_HE_SE_nat	-0.7215029	1.97194304
z_SE_EI_nat	-0.9909774	6.46029686
z_SE_EO_nat	-4.9462912	0.02654865
z_SE_ER_nat	-3.1792654	1.91290954
z_SE_HQ_nat	0.1598976	4.85756804
z_SE_SR_nat	-3.3206845	1.15396296
z_SE_WL_nat	-2.4104961	2.96592973

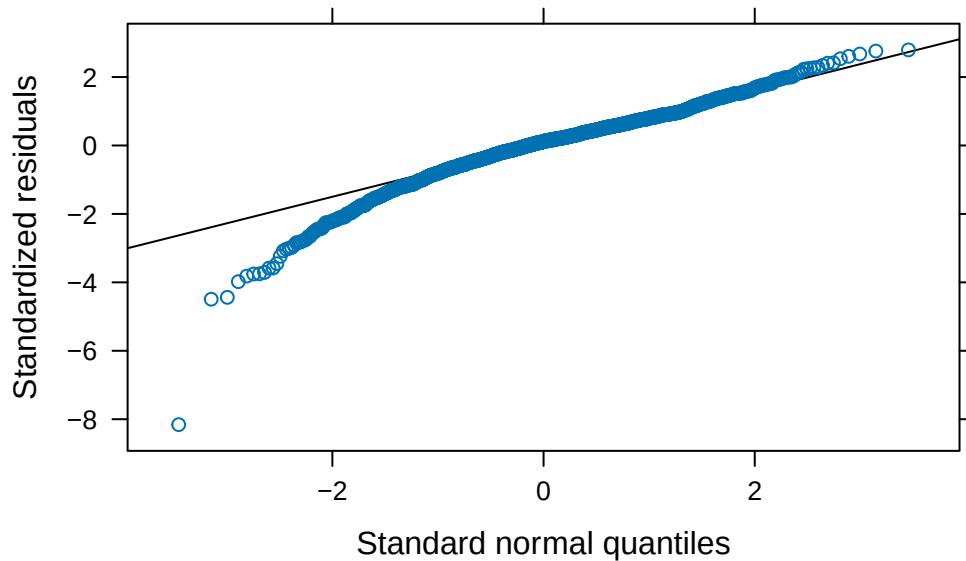
```
plot(mod3_eff, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_eff,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_eff)
```



```
#Table
#fixed effects
mod3_eff_table <- data.frame(
  predictor = attr(summary(mod3_eff)$coef,"dimnames")[[1]],
  B = summary(mod3_eff)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod3_eff)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_eff)$CI_low,4),", ",round(
  pval = summary(mod3_eff)$coef[,5]
)

mod3_eff_table[,c("B","beta","pval")] <-
round(mod3_eff_table[,c("B","beta","pval")],4)
write.csv(mod3_eff_table,file = "../bohnert_sleep_git/mod3_eff_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_eff)$sigma^2
#between
tau00 <- attr(summary(mod3_eff)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_eff)$devcomp$dims["q"]
obs <- summary(mod3_eff)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_eff)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_eff)[,"R2c"]
mod3_eff_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
```



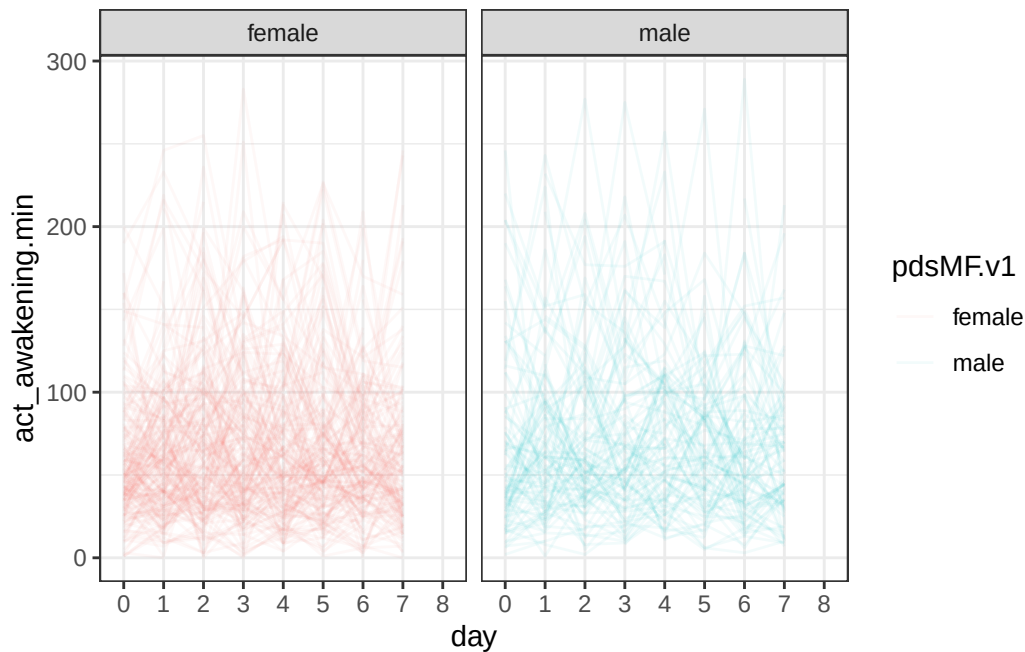
```
mod3_eff_table_re
```

```
          sigma2    tau00      ICC    N obs      R2marg    R2cond  
(Intercept) 46.72389 38.39828 0.4510961 255 1821 0.07668618 0.4931894
```

Awakening

```
ggplot(aes(x = day, y = act_awakening.min, group = ID, color = pdsMF.v1), data = datlong %>%
```

Warning: Removed 450 rows containing missing values or values outside the scale range (`geom\_line()`).



```
library(lme4)
#####
#act_awakening.min
#####
#####
#Model 1: COI Total
#####
mod1_awake <- lmer(act_awakening.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp
summary(mod1_awake)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_awakening.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_COI_nat + (1 | ID)
Data: datlong

REML criterion at convergence: 21458.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.6768	-0.5852	-0.1524	0.4399	5.1610

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	834	28.88
Residual		1326	36.41

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	52.1439	10.1307	283.7173	5.147	4.95e-07 ***
pdsMF.v1male	5.1499	3.9963	280.1834	1.289	0.19857
cg1.age.v1	0.4685	0.2005	281.1148	2.337	0.02015 *
income.v1.10000	-1.7825	0.6560	277.4306	-2.717	0.00699 **
cg1_edu_grp<=BS	-1.6893	4.4282	285.6162	-0.381	0.70313
cg1_edu_grp>=MS	2.3063	7.6029	281.9641	0.303	0.76185
I_frisat	4.6509	1.7971	1817.7178	2.588	0.00973 **
z_COI_nat	1.1692	2.6469	283.8656	0.442	0.65903

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst
pdsMF.v1mal	-0.124						
cg1.age.v1	-0.886	0.015					
inc.1.10000	-0.205	0.011	-0.021				
cg1_d_g<=BS	-0.265	-0.069	-0.002	-0.165			
cg1_d_g>=MS	-0.066	-0.141	-0.052	-0.304	0.469		
I_frisat	-0.045	0.004	-0.003	0.007	-0.004	-0.012	
z_COI_nat	0.216	-0.014	-0.039	-0.141	-0.074	-0.110	0.007

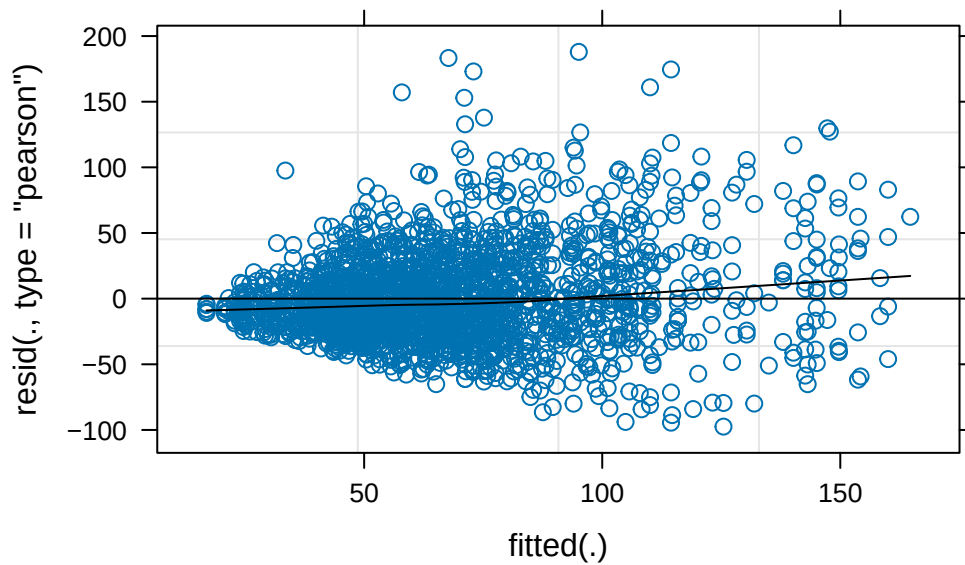
```
ci <- confint(mod1_awake)
```

Computing profile confidence intervals ...

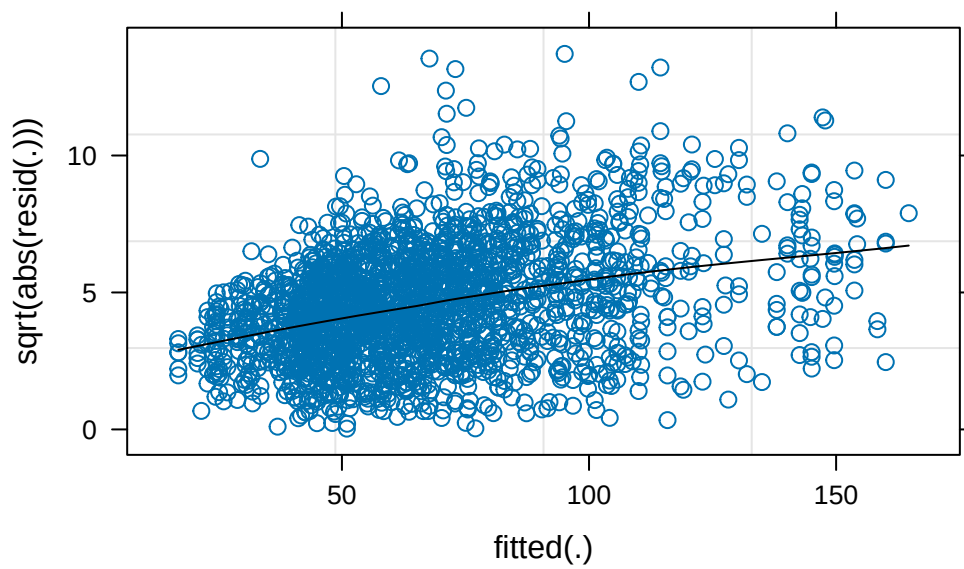
```
ci
```

	2.5 %	97.5 %
.sig01	25.72697443	31.4970820
.sigma	35.24579754	37.6242867
(Intercept)	32.47274441	71.8319307
pdsMF.v1male	-2.61147381	12.9137917
cg1.age.v1	0.07885699	0.8578054
income.v1.10000	-3.05673127	-0.5084468
cg1_edu_grp<=BS	-10.29156667	6.9123478
cg1_edu_grp>=MS	-12.46300144	17.0739437
I_frisat	1.13136583	8.1774300
z_COI_nat	-3.97403489	6.3092256

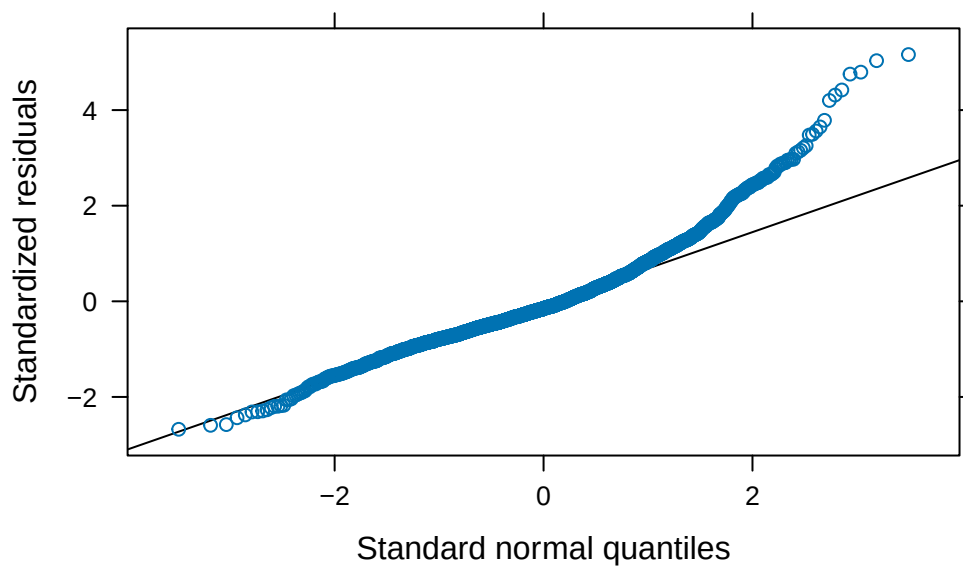
```
plot(mod1_awake, type=c("p", "smooth"), col.line=1)
```



```
plot(mod1_awake,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p", "smooth"), col.line=1)
```



```
lattice::qqmath(mod1_awake)
```



```
#Table
#Fixed effects
mod1_awake_table <- data.frame(
  predictor = attr(summary(mod1_awake)$coef,"dimnames")[[1]],
  B = summary(mod1_awake)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod1_awake)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effects::standardize_parameters(mod1_awake)$CI_low,4),", ",round
pval = summary(mod1_awake)$coef[,5]
)

mod1_awake_table[,c("B","beta","pval")] <-
round(mod1_awake_table[,c("B","beta","pval")],4)
write.csv(mod1_awake_table,file = "../bohnert_sleep_git/mod1_awake_table.csv")

#Random awakeects
#within
sigma2 <- summary(mod1_awake)$sigma^2
#between
tau00 <- attr(summary(mod1_awake)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_awake)$devcomp$dims["q"]
obs <- summary(mod1_awake)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_awake)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_awake)[,"R2c"]
mod1_awake_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_awake_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	1325.854	833.9885	0.3861339	291	2094	0.02565152	0.4018805

```

#####
#Model 2: COI Sub-domains
#####
mod2_awake <- lmer(act_awakening.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp
summary(mod2_awake)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: act_awakening.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)

Data: datlong

REML criterion at convergence: 21448.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.6871	-0.5823	-0.1507	0.4413	5.1571

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	838	28.95
Residual		1326	36.41

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	52.5569	10.4499	281.6263	5.029	8.77e-07 ***
pdsMF.v1male	5.4330	4.0146	278.3859	1.353	0.17706
cg1.age.v1	0.4630	0.2013	279.6643	2.300	0.02218 *
income.v1.10000	-1.6869	0.6660	276.5838	-2.533	0.01187 *
cg1_edu_grp<=BS	-1.4128	4.4450	284.0643	-0.318	0.75084
cg1_edu_grp>=MS	2.3002	7.6220	280.3826	0.302	0.76305
I_frisat	4.6519	1.7970	1818.0836	2.589	0.00971 **
z_ED_nat	5.3766	4.7977	285.9944	1.121	0.26337
z_HE_nat	-2.0046	4.5373	279.0302	-0.442	0.65897
z_SE_nat	-2.1568	3.9994	281.8823	-0.539	0.59012

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
pdsMF.v1mal	-0.111								
cg1.age.v1	-0.873	0.011							
inc.1.10000	-0.213	0.017	-0.018						
cg1_d_g<=BS	-0.255	-0.065	-0.003	-0.154					
cg1_d_g>=MS	-0.072	-0.142	-0.050	-0.297	0.468				
I_frisat	-0.045	0.004	-0.003	0.007	-0.004	-0.012			
z_ED_nat	0.015	0.055	-0.018	0.128	0.047	-0.008	0.002		
z_HE_nat	-0.203	-0.058	0.061	0.004	-0.033	0.019	0.004	-0.241	
z_SE_nat	0.212	-0.028	-0.038	-0.198	-0.067	-0.064	0.000	-0.711	-0.258

```
ci <- confint(mod2_awake)
```

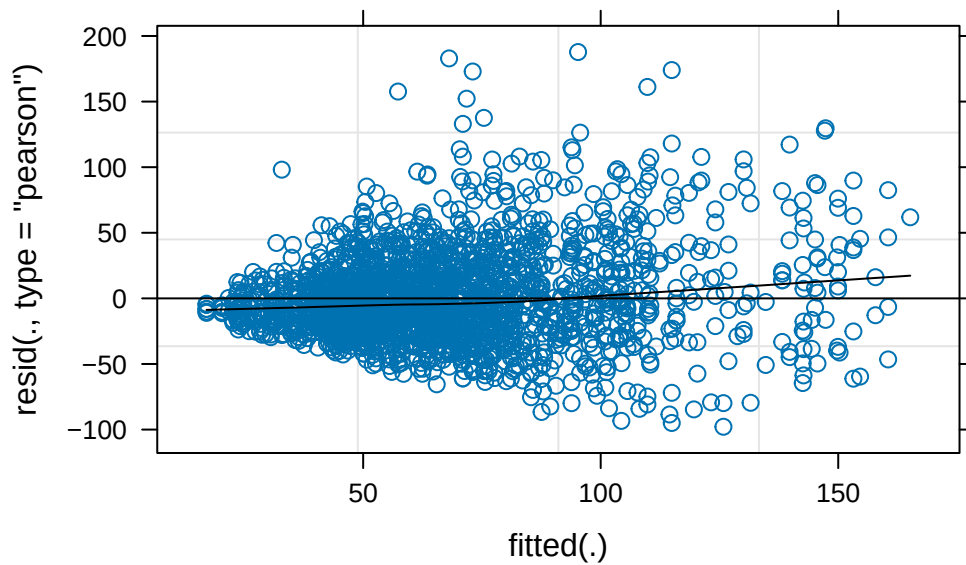
Computing profile confidence intervals ...

```
ci
```

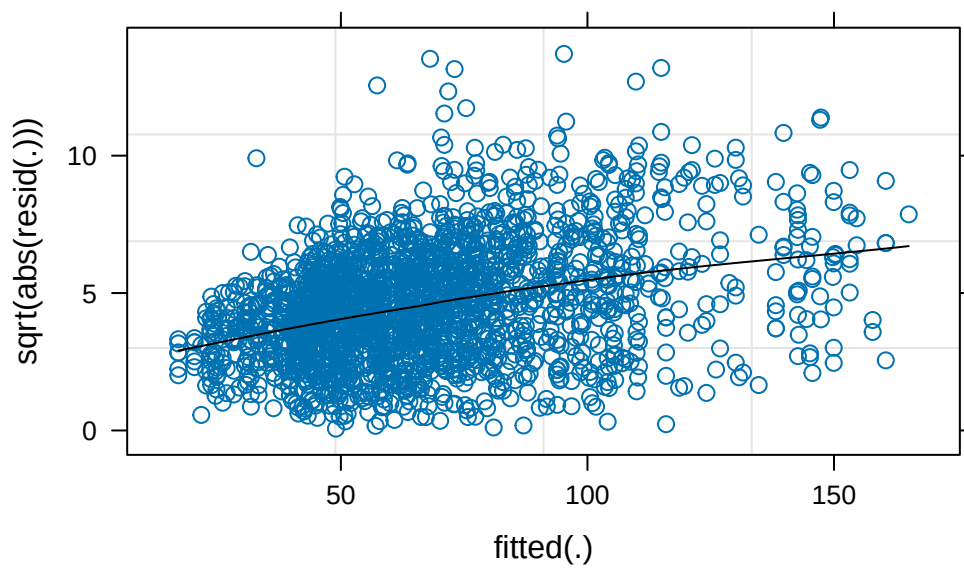
2.5 % 97.5 %

```
.sig01          25.67769121 31.4353831
.sigma          35.24295901 37.6209052
(Intercept)     32.34448452 72.8010164
pdsMF.v1male    -2.33646097 13.2048780
cg1.age.v1       0.07301936  0.8523753
income.v1.10000 -2.97639220 -0.3981916
cg1_edu_grp<=BS -10.01849039  7.1904247
cg1_edu_grp>=MS -12.45509155 17.0519064
I_frisat         1.13367576  8.1791489
z_ED_nat        -3.92386840 14.6524667
z_HE_nat        -10.78517892  6.7796363
z_SE_nat        -9.89208720  5.5916922
```

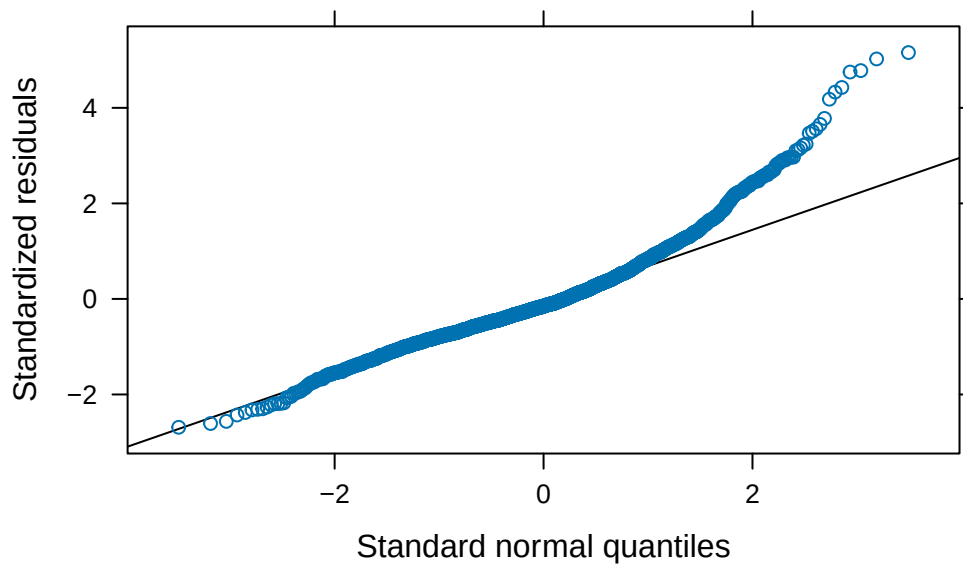
```
plot(mod2_awake, type=c("p","smooth"), col.line=1)
```



```
plot(mod2_awake,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_awake)
```



```
#Table
#Fixed effects
mod2_awake_table <- data.frame(
  predictor = attr(summary(mod2_awake)$coef,"dimnames")[[1]],
  B = summary(mod2_awake)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod2_awake)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```



```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_awake)$CI_low,4),", ",round
pval = summary(mod2_awake)$coef[,5]
)

mod2_awake_table[,c("B","beta","pval")] <-
round(mod2_awake_table[,c("B","beta","pval")],4)
write.csv(mod2_awake_table,file = "../bohnert_sleep_git/mod2_awake_table.csv")

#Random awakeects
#within
sigma2 <- summary(mod2_awake)$sigma^2
#between
tau00 <- attr(summary(mod2_awake)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_awake)$devcomp$dims["q"]
obs <- summary(mod2_awake)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_awake)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_awake)[,"R2c"]
mod2_awake_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_awake_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	1325.637	837.9963	0.3873098	291	2094	0.02762918	0.4042379

```

#####
#Model 3: COI Sub-domains
#####
mod3_awake <- lmer(act_awakening.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_awake)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: act_awakening.min ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +
cg1_edu_grp + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat +
(1 | ID)

Data: datlong

REML criterion at convergence: 18673.9

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.6405	-0.5853	-0.1652	0.4221	5.1155

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	854.9	29.24
	Residual	1401.9	37.44

Number of obs: 1821, groups: ID, 255

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	62.31718	12.22679	233.36291	5.097	7.14e-07 ***
pdsMF.v1male	4.23729	4.43939	232.06125	0.954	0.34084
cg1.age.v1	0.43815	0.21620	233.14057	2.027	0.04385 *
income.v1.10000	-2.18700	0.73963	230.31992	-2.957	0.00343 **
cg1_edu_grp<=BS	2.25179	4.98052	236.37522	0.452	0.65160
cg1_edu_grp>=MS	7.48285	8.40393	233.96543	0.890	0.37417
z_ED_EC_nat	0.68423	2.46301	231.06128	0.278	0.78141
z_ED_EL_nat	6.96888	5.56281	236.37628	1.253	0.21153
z_ED_ER_nat	5.22214	9.62060	234.88593	0.543	0.58778
z_ED_SP_nat	0.02979	5.26372	234.62359	0.006	0.99549
z_HE_EP_nat	-32.07593	21.48277	233.27789	-1.493	0.13676
z_HE_HE_nat	-4.19765	12.15628	235.34838	-0.345	0.73017
z_HE_HR_nat	8.17060	4.99385	232.58882	1.636	0.10316
z_HE_SE_nat	-1.07385	3.45307	237.05452	-0.311	0.75609
z_SE_EI_nat	-15.24987	9.54910	235.00542	-1.597	0.11161
z_SE_EO_nat	7.67828	6.37309	235.19810	1.205	0.22949
z_SE_ER_nat	3.57770	6.52555	234.78705	0.548	0.58403
z_SE_HQ_nat	-15.30816	6.01820	233.10817	-2.544	0.01162 *
z_SE_SR_nat	3.11088	5.73234	233.01229	0.543	0.58786
z_SE_WL_nat	1.35388	6.88751	233.05133	0.197	0.84434

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation matrix not shown by default, as $p = 20 > 12$.

Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

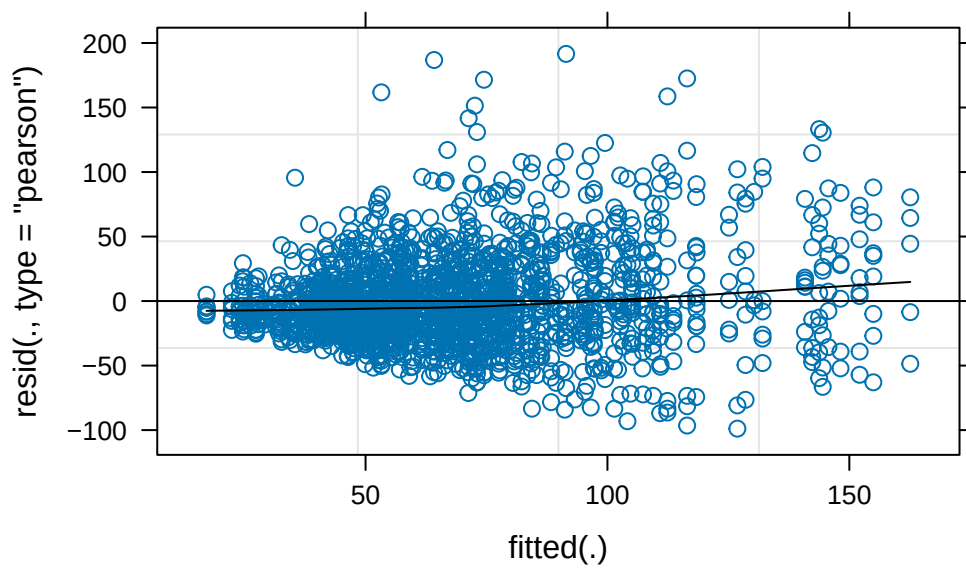
```
ci <- confint(mod3_awake)
```

Computing profile confidence intervals ...

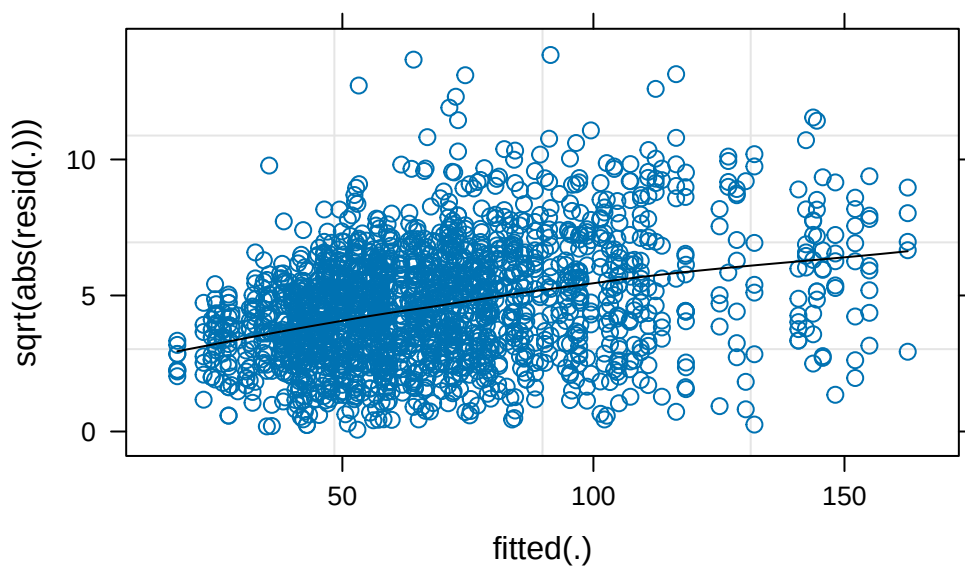
```
ci
```

	2.5 %	97.5 %
.sig01	24.88378308	31.0384086
.sigma	36.16871164	38.7933012
(Intercept)	39.29796707	85.4737319
pdsMF.v1male	-4.13933342	12.6243882
cg1.age.v1	0.02822434	0.8447545
income.v1.10000	-3.58489936	-0.7923203
cg1_edu_grp<=BS	-7.16856916	11.6445256
cg1_edu_grp>=MS	-8.39477728	23.3439942
z_ED_EC_nat	-3.96588029	5.3340413
z_ED_EL_nat	-3.55961734	17.4532495
z_ED_ER_nat	-12.95852722	23.3775980
z_ED_SP_nat	-9.94411171	9.9369958
z_HE_EP_nat	-72.57339532	8.5567186
z_HE_HE_nat	-27.17100246	18.7437313
z_HE_HR_nat	-1.26553616	17.5926644
z_HE_SE_nat	-7.58482569	5.4592647
z_SE_EI_nat	-33.24588937	2.8210391
z_SE_EO_nat	-4.34442733	19.7268175
z_SE_ER_nat	-8.77230549	15.8743659
z_SE_HQ_nat	-26.67976007	-3.9525323
z_SE_SR_nat	-7.72282302	13.9248027
z_SE_WL_nat	-11.61406636	14.3968321

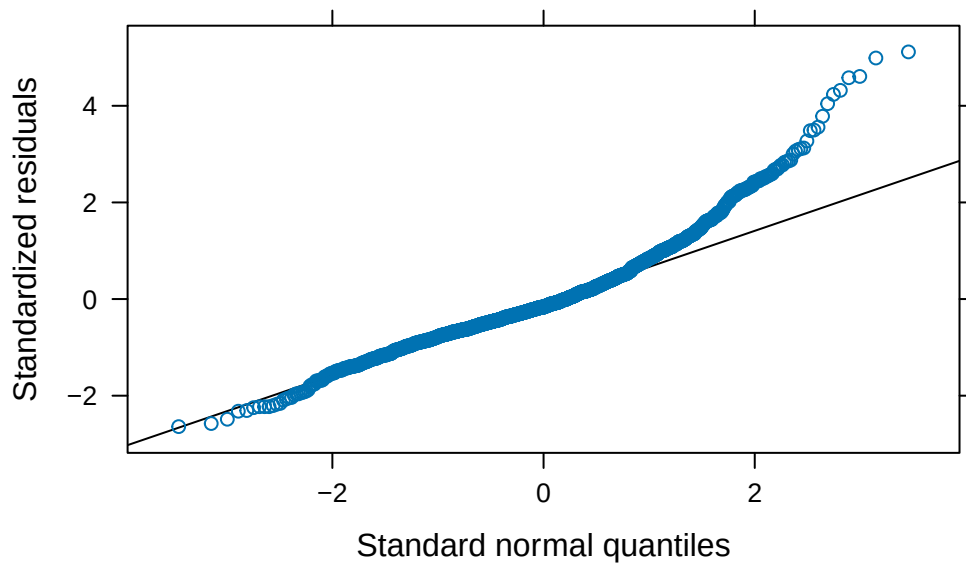
```
plot(mod3_awake, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_awake,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_awake)
```



```
#Table
#Fixed effects
mod3_awake_table <- data.frame(
  predictor = attr(summary(mod3_awake)$coef,"dimnames")[[1]],
  B = summary(mod3_awake)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod3_awake)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_awake)$CI_low,4),", ",round(
  pval = summary(mod3_awake)$coef[,5]
)

mod3_awake_table[,c("B","beta","pval")] <-
round(mod3_awake_table[,c("B","beta","pval")],4)
write.csv(mod3_awake_table,file = "../bohnert_sleep_git/mod3_awake_table.csv")

#Random awakeects
#within
sigma2 <- summary(mod3_awake)$sigma^2
#between
tau00 <- attr(summary(mod3_awake)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_awake)$devcomp$dims["q"]
obs <- summary(mod3_awake)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_awake)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_awake)[,"R2c"]
mod3_awake_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
```

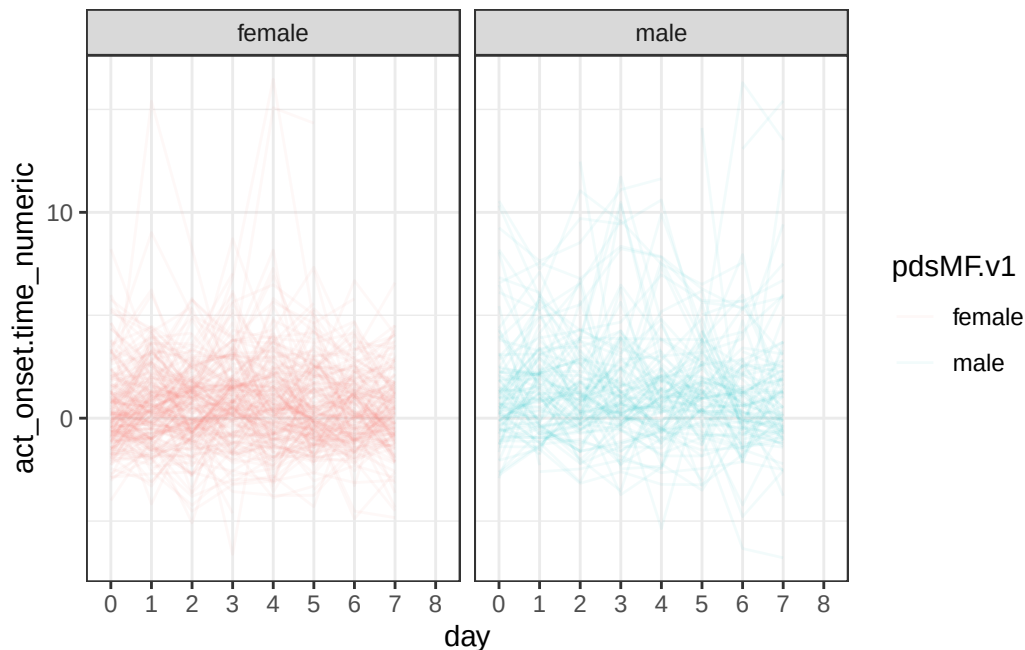
```
mod3_awake_table_re
```

```
          sigma2    tau00      ICC    N obs      R2marg    R2cond  
(Intercept) 1401.918 854.9311 0.3788162 255 1821 0.05724614 0.4143766
```

Sleep Onset

```
ggplot(aes(x = day, y = act_onset.time_numeric, group = ID, color = pdsMF.v1), data = datlong)
```

Warning: Removed 450 rows containing missing values or values outside the scale range (`geom\_line()`).



```
library(lme4)
#####
#act_onset.time_numeric
#####
#####
#Model 1: COI Total
#####
mod1_onset <- lmer(act_onset.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu
summary(mod1_onset)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
 lmerModLmerTest]
 Formula: act_onset.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +
 cg1_edu_grp + I_frisat + z_COI_nat + (1 | ID)
 Data: datlong

REML criterion at convergence: 8879

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.4551	-0.5024	-0.0473	0.4521	7.1401

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.960	1.721
	Residual	3.031	1.741

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	8.034e-01	5.828e-01	2.876e+02	1.378	0.16913
pdsMF.v1male	7.615e-01	2.302e-01	2.845e+02	3.309	0.00106 **
cg1.age.v1	-5.833e-03	1.154e-02	2.853e+02	-0.505	0.61374
income.v1.10000	-4.058e-03	3.781e-02	2.823e+02	-0.107	0.91460
cg1_edu_grp<=BS	9.722e-02	2.546e-01	2.891e+02	0.382	0.70285
cg1_edu_grp>=MS	4.003e-01	4.376e-01	2.858e+02	0.915	0.36111
I_frisat	6.244e-01	8.598e-02	1.816e+03	7.262	5.65e-13 ***
z_COI_nat	1.965e-01	1.523e-01	2.873e+02	1.290	0.19794

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst
pdsMF.v1mal	-0.124						
cg1.age.v1	-0.886	0.015					
inc.1.10000	-0.206	0.010	-0.020				
cg1_d_g<=BS	-0.263	-0.071	-0.002	-0.165			
cg1_d_g>=MS	-0.064	-0.142	-0.053	-0.305	0.468		
I_frisat	-0.038	0.003	-0.003	0.006	-0.004	-0.010	
z_COI_nat	0.215	-0.013	-0.039	-0.142	-0.072	-0.107	0.006

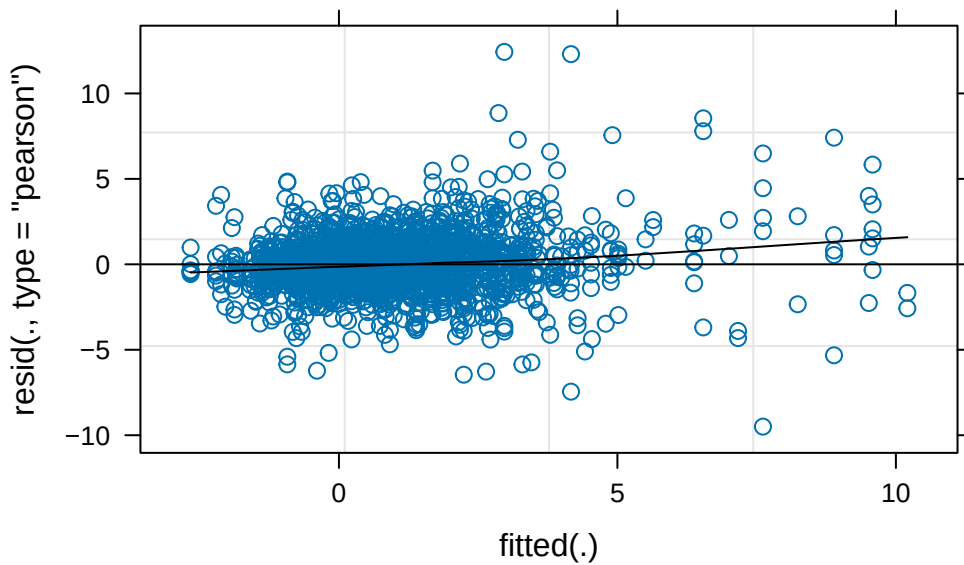
```
ci <- confint(mod1_onset)
```

Computing profile confidence intervals ...

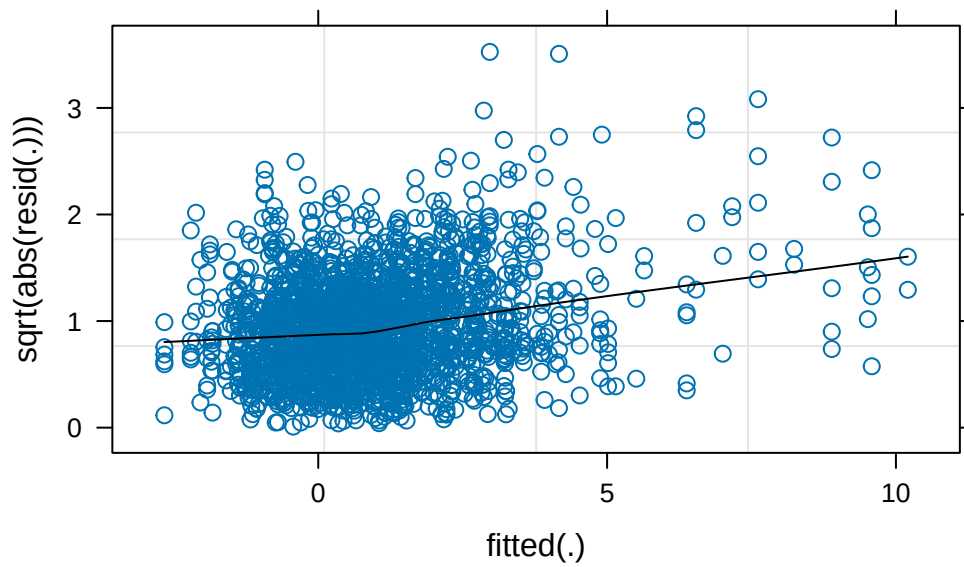
```
ci
```

	2.5 %	97.5 %
.sig01	1.54736440	1.86566698
.sigma	1.68511236	1.79871386
(Intercept)	-0.32933208	1.93527828
pdsMF.v1male	0.31441568	1.20867829
cg1.age.v1	-0.02825587	0.01659786
income.v1.10000	-0.07750115	0.06940156
cg1_edu_grp<=BS	-0.39732995	0.59199947
cg1_edu_grp>=MS	-0.44974568	1.25071858
I_frisat	0.45578789	0.79289795
z_COI_nat	-0.09944383	0.49224564

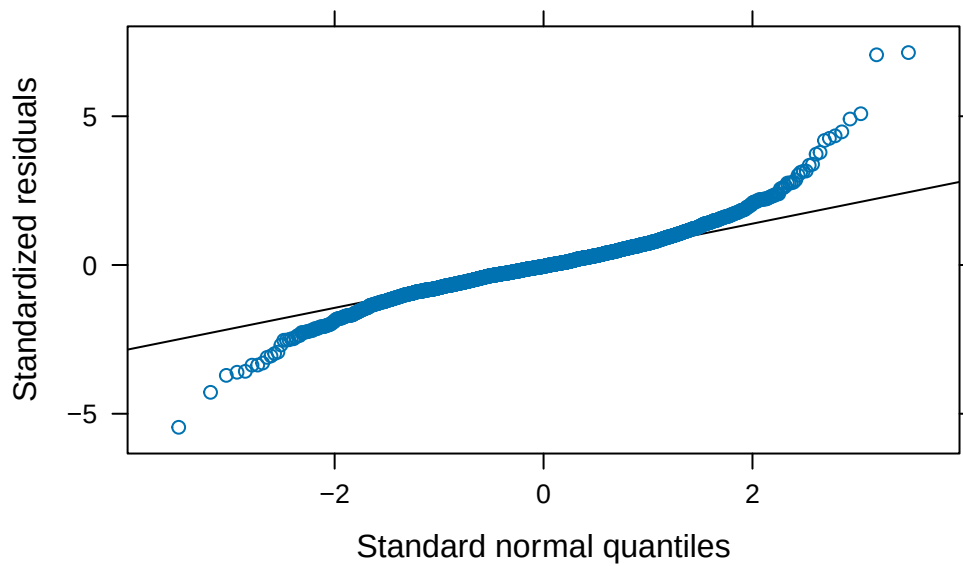
```
plot(mod1_onset, type=c("p","smooth"), col.line=1)
```



```
plot(mod1_onset,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```

```
lattice::qqmath(mod1_onset)
```



```
#Table
#Fixed effects
mod1_onset_table <- data.frame(
  predictor = attr(summary(mod1_onset)$coef,"dimnames")[[1]],
  B = summary(mod1_onset)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod1_onset)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_onset)$CI_low,4),", ",round
pval = summary(mod1_onset)$coef[,5]
)

mod1_onset_table[,c("B","beta","pval")] <-
round(mod1_onset_table[,c("B","beta","pval")],4)
write.csv(mod1_onset_table,file = "../bohnert_sleep_git/mod1_onset_table.csv")

#Random effects
#within
sigma2 <- summary(mod1_onset)$sigma^2
#between
tau00 <- attr(summary(mod1_onset)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_onset)$devcomp$dims["q"]
obs <- summary(mod1_onset)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_onset)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_onset)[,"R2c"]
mod1_onset_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_onset_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	3.03064	2.960121	0.4941144	291	2094	0.04166761	0.5151934

```

#####
#Model 2: COI Sub-domains
#####
mod2_onset <- lmer(act_onset.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu
summary(mod2_onset)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: act_onset.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)
Data: datlong

REML criterion at convergence: 8880.4

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.4629	-0.5032	-0.0443	0.4511	7.1356

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.975	1.725
	Residual	3.031	1.741

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	7.760e-01	6.014e-01	2.850e+02	1.290	0.19802
pdsMF.v1male	7.486e-01	2.313e-01	2.820e+02	3.236	0.00135 **
cg1.age.v1	-5.557e-03	1.159e-02	2.831e+02	-0.479	0.63206
income.v1.10000	-7.719e-03	3.839e-02	2.806e+02	-0.201	0.84081
cg1_edu_grp<=BS	8.484e-02	2.557e-01	2.868e+02	0.332	0.74025
cg1_edu_grp>=MS	4.007e-01	4.389e-01	2.836e+02	0.913	0.36204
I_frisat	6.244e-01	8.599e-02	1.816e+03	7.262	5.65e-13 ***
z_ED_nat	-1.441e-01	2.758e-01	2.881e+02	-0.523	0.60170
z_HE_nat	1.489e-01	2.613e-01	2.831e+02	0.570	0.56930
z_SE_nat	2.082e-01	2.302e-01	2.849e+02	0.904	0.36652

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
pdsMF.v1male	-0.111								
cg1.age.v1	-0.874	0.012							
inc.1.10000	-0.214	0.016	-0.017						
cg1_d_g<=BS	-0.254	-0.066	-0.003	-0.154					
cg1_d_g>=MS	-0.070	-0.142	-0.051	-0.298	0.467				
I_frisat	-0.037	0.003	-0.002	0.006	-0.004	-0.010			
z_ED_nat	0.013	0.056	-0.016	0.127	0.047	-0.007	0.002		
z_HE_nat	-0.203	-0.059	0.060	0.004	-0.032	0.020	0.003	-0.241	
z_SE_nat	0.213	-0.028	-0.039	-0.198	-0.067	-0.064	0.000	-0.710	-0.259

```
ci <- confint(mod2_onset)
```

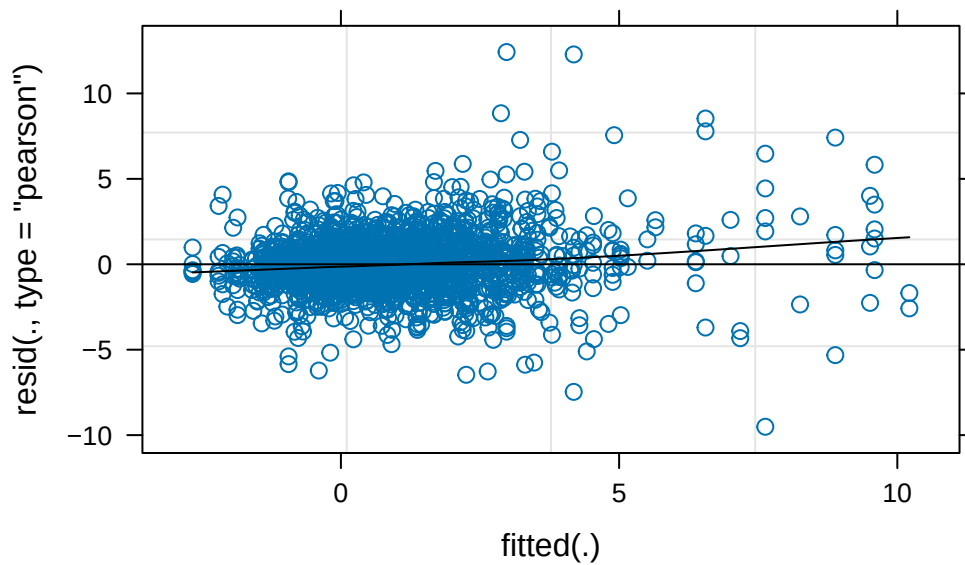
Computing profile confidence intervals ...

```
ci
```

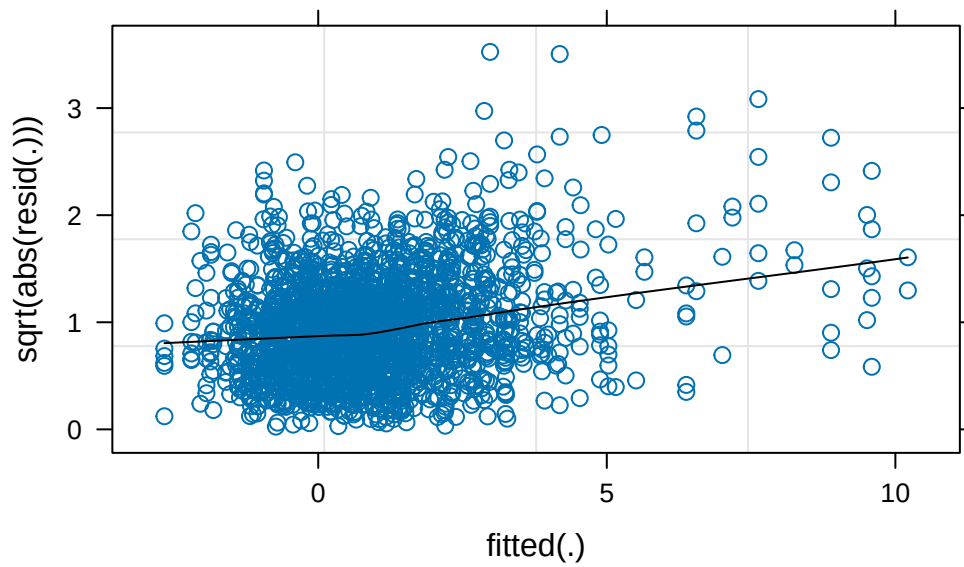
2.5 % 97.5 %

```
.sig01      1.54490290 1.86295628
.sigma      1.68515889 1.79876937
(Intercept) -0.38915187 1.93961633
pdsMF.v1male 0.30086849 1.19639827
cg1.age.v1   -0.02799430 0.01689495
income.v1.10000 -0.08203242 0.06660966
cg1_edu_grp<=BS -0.40995196 0.57995553
cg1_edu_grp>=MS -0.44868257 1.25058219
I_frisat     0.45579583 0.79291602
z_ED_nat     -0.67846726 0.38946850
z_HE_nat     -0.35691285 0.65491896
z_SE_nat     -0.23733402 0.65390778
```

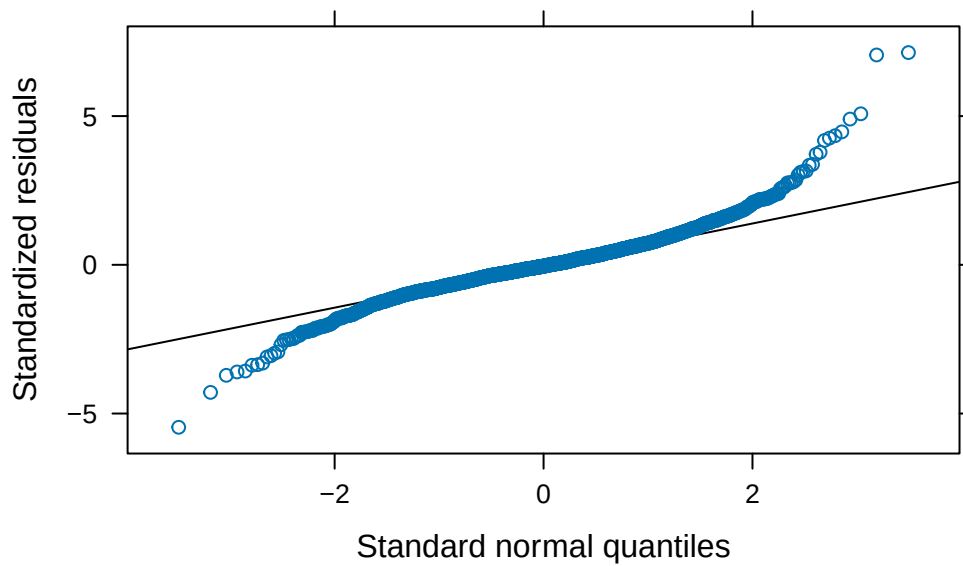
```
plot(mod2_onset, type=c("p","smooth"), col.line=1)
```



```
plot(mod2_onset,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_onset)
```



```
#Table
#Fixed effects
mod2_onset_table <- data.frame(
  predictor = attr(summary(mod2_onset)$coef,"dimnames")[[1]],
  B = summary(mod2_onset)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod2_onset)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_onset)$CI_low,4),", ",round
pval = summary(mod2_onset)$coef[,5]
)

mod2_onset_table[,c("B","beta","pval")] <-
round(mod2_onset_table[,c("B","beta","pval")],4)
write.csv(mod2_onset_table,file = "../bohnert_sleep_git/mod2_onset_table.csv")

#Random effects
#within
sigma2 <- summary(mod2_onset)$sigma^2
#between
tau00 <- attr(summary(mod2_onset)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_onset)$devcomp$dims["q"]
obs <- summary(mod2_onset)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_onset)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_onset)[,"R2c"]
mod2_onset_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_onset_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	3.030843	2.975066	0.4953565	291	2094	0.04266245	0.5168858

```

#####
#Model 3: COI Sub-domains
#####
mod3_onset <- lmer(act_onset.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_edu
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_onset)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_onset.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +
      cg1_edu_grp + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
      z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
      z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat +
      (1 | ID)
Data: datlong

```

REML criterion at convergence: 7696.4

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.7011	-0.5022	-0.0673	0.4843	7.1250

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	3.078	1.754
	Residual	2.982	1.727

Number of obs: 1821, groups: ID, 255

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.260842	0.704128	237.388717	0.370	0.71138
pdsMF.v1male	0.839360	0.255833	235.888733	3.281	0.00119 **
cg1.age.v1	-0.002316	0.012453	236.826844	-0.186	0.85261
income.v1.10000	-0.024511	0.042654	234.582248	-0.575	0.56608
cg1_edu_grp<=BS	0.101198	0.286479	239.517915	0.353	0.72421
cg1_edu_grp>=MS	0.439119	0.483921	237.309677	0.907	0.36511
z_ED_EC_nat	-0.333267	0.141980	235.592107	-2.347	0.01974 *
z_ED_EL_nat	0.243200	0.320000	239.192453	0.760	0.44800
z_ED_ER_nat	-0.715713	0.553731	238.266235	-1.293	0.19743
z_ED_SP_nat	0.204589	0.303006	237.946526	0.675	0.50021
z_HE_EP_nat	0.921484	1.237505	236.452970	0.745	0.45723
z_HE_HE_nat	0.480279	0.699568	238.486286	0.687	0.49304
z_HE_HR_nat	0.021870	0.287720	236.310413	0.076	0.93947
z_HE_SE_nat	-0.077220	0.198561	240.119201	-0.389	0.69770
z_SE_EI_nat	0.917992	0.549598	238.288269	1.670	0.09617 .
z_SE_EO_nat	-0.095027	0.366791	238.267833	-0.259	0.79580
z_SE_ER_nat	-0.441132	0.375611	238.135381	-1.174	0.24139
z_SE_HQ_nat	0.495543	0.346648	236.861508	1.430	0.15417
z_SE_SR_nat	-0.115640	0.330201	236.737636	-0.350	0.72649
z_SE_WL_nat	0.174107	0.396751	236.623779	0.439	0.66118

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation matrix not shown by default, as $p = 20 > 12$.

Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

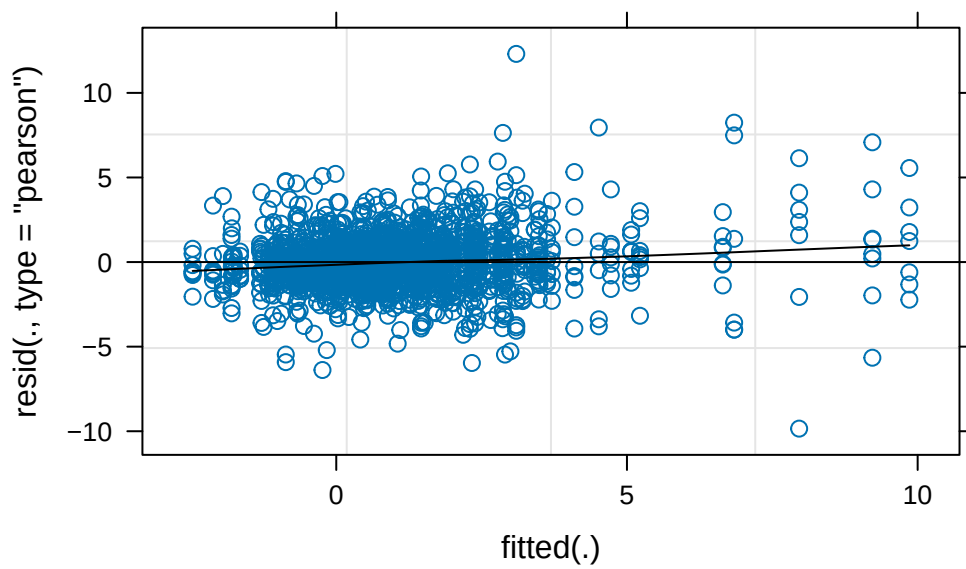
```
ci <- confint(mod3_onset)
```

Computing profile confidence intervals ...

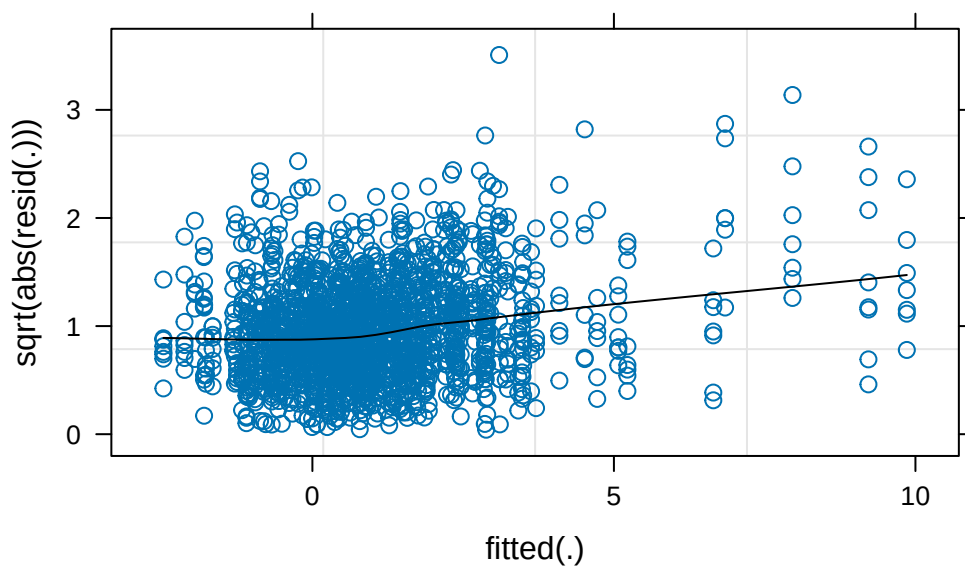
```
ci
```

	2.5 %	97.5 %
.sig01	1.51777527	1.85397589
.sigma	1.66802079	1.78893353
(Intercept)	-1.07208392	1.58867813
pdsMF.v1male	0.35615521	1.32278126
cg1.age.v1	-0.02581481	0.02124188
income.v1.10000	-0.10502235	0.05612524
cg1_edu_grp<=BS	-0.43984290	0.64285535
cg1_edu_grp>=MS	-0.47426103	1.35435608
z_ED_EC_nat	-0.60159661	-0.06515744
z_ED_EL_nat	-0.36191851	0.84743944
z_ED_ER_nat	-1.76362544	0.32893786
z_ED_SP_nat	-0.36873541	0.77630266
z_HE_EP_nat	-1.41789429	3.25803809
z_HE_HE_nat	-0.84154685	1.80215543
z_HE_HR_nat	-0.52176630	0.56537521
z_HE_SE_nat	-0.45175613	0.29871223
z_SE_EI_nat	-0.12002198	1.95690781
z_SE_EO_nat	-0.78756192	0.59854208
z_SE_ER_nat	-1.15191928	0.26751102
z_SE_HQ_nat	-0.15920458	1.15064460
z_SE_SR_nat	-0.73842942	0.50927650
z_SE_WL_nat	-0.57496199	0.92418439

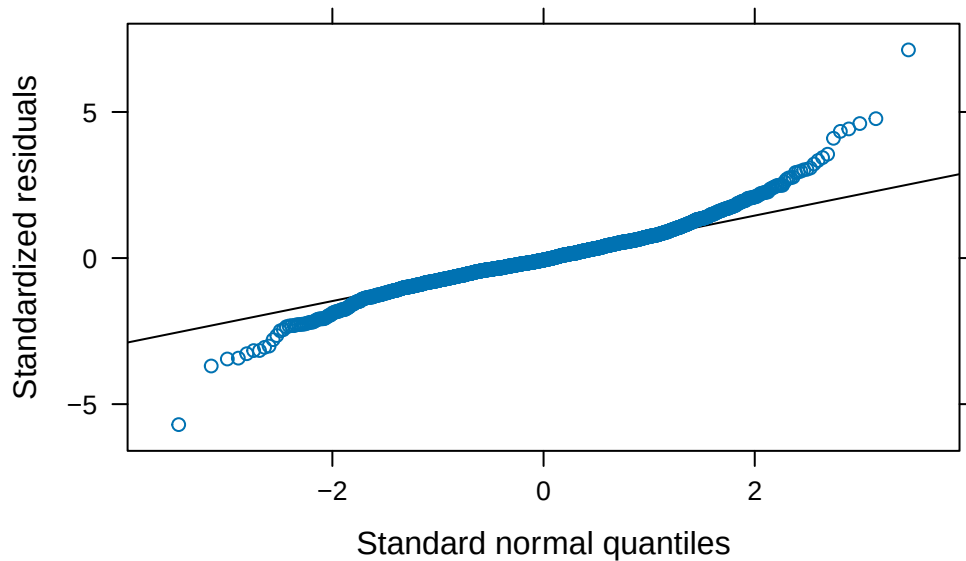
```
plot(mod3_onset, type=c("p","smooth"), col.line=1)
```

```
plot(mod3_onset,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_onset)
```



```
#Table
#fixed effects
mod3_onset_table <- data.frame(
  predictor = attr(summary(mod3_onset)$coef,"dimnames")[[1]],
  B = summary(mod3_onset)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod3_onset)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_onset)$CI_low,4),", ",round(
  pval = summary(mod3_onset)$coef[,5]
)

mod3_onset_table[,c("B","beta","pval")] <-
round(mod3_onset_table[,c("B","beta","pval")],4)
write.csv(mod3_onset_table,file = "../bohnert_sleep_git/mod3_onset_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_onset)$sigma^2
#between
tau00 <- attr(summary(mod3_onset)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_onset)$devcomp$dims["q"]
obs <- summary(mod3_onset)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_onset)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_onset)[,"R2c"]
mod3_onset_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
```

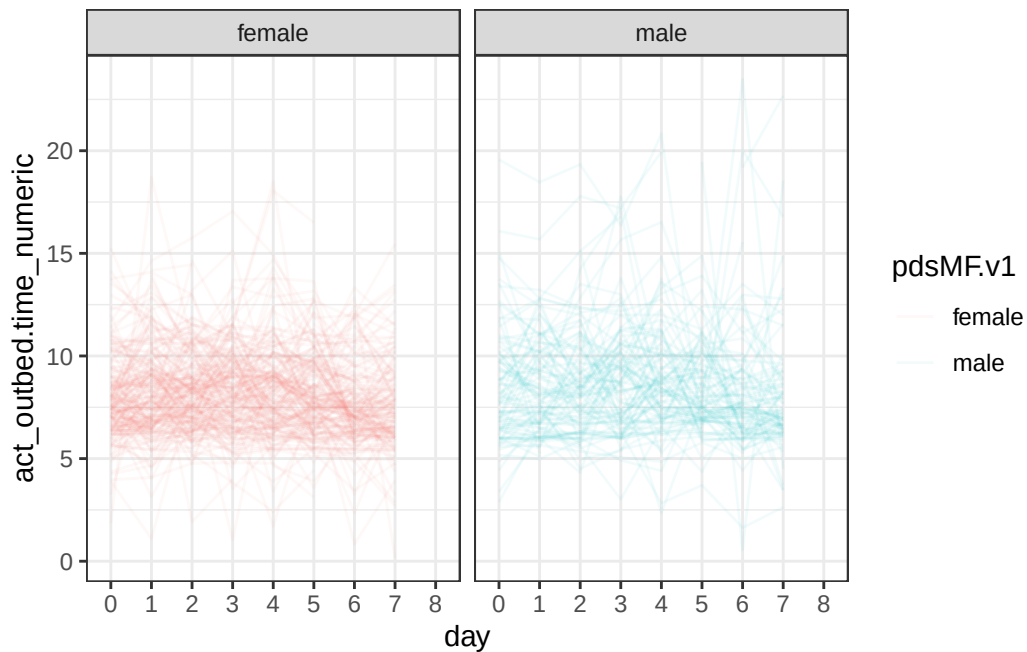
```
mod3_onset_table_re
```

```
          sigma2    tau00      ICC    N  obs      R2marg    R2cond  
(Intercept) 2.981984 3.078175 0.5079364 255 1821 0.05831472 0.5366309
```

Sleep Outbed

```
ggplot(aes(x = day, y = act_outbed.time_numeric, group = ID, color = pdsMF.v1), data = datlon)
```

Warning: Removed 450 rows containing missing values or values outside the scale range (`geom\_line()`).



```
library(lme4)  
#####  
#act_outbed.time_numeric  
#####  
#####  
#Model 1: COI Total  
#####  
mod1_outbed <- lmer(act_outbed.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_  
summary(mod1_outbed)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
 lmerModLmerTest]
 Formula: act_outbed.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +
 cg1_edu_grp + I_frisat + z_COI_nat + (1 | ID)
 Data: datlong

REML criterion at convergence: 8849

Scaled residuals:

Min	1Q	Median	3Q	Max
-7.1194	-0.4729	-0.0510	0.4509	8.7373

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.761	1.662
	Residual	3.009	1.735

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	8.61027	0.56526	284.30636	15.232	<2e-16 ***
pdsMF.v1male	0.40602	0.22319	281.12283	1.819	0.070 .
cg1.age.v1	-0.01103	0.01119	281.92832	-0.986	0.325
income.v1.10000	-0.03214	0.03666	278.91876	-0.877	0.381
cg1_edu_grp<=BS	-0.11788	0.24696	285.83659	-0.477	0.633
cg1_edu_grp>=MS	0.27988	0.42443	282.55498	0.659	0.510
I_frisat	1.24033	0.08567	1813.88444	14.478	<2e-16 ***
z_COI_nat	0.24045	0.14769	284.06187	1.628	0.105

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1..1	i.1.10	c1__<=	c1__>=	I_frst
pdsMF.v1mal	-0.124						
cg1.age.v1	-0.886	0.015					
inc.1.10000	-0.206	0.010	-0.020				
cg1_d_g<=BS	-0.263	-0.070	-0.002	-0.165			
cg1_d_g>=MS	-0.064	-0.142	-0.053	-0.305	0.468		
I_frisat	-0.039	0.003	-0.003	0.006	-0.004	-0.010	
z_COI_nat	0.215	-0.013	-0.039	-0.142	-0.072	-0.108	0.006

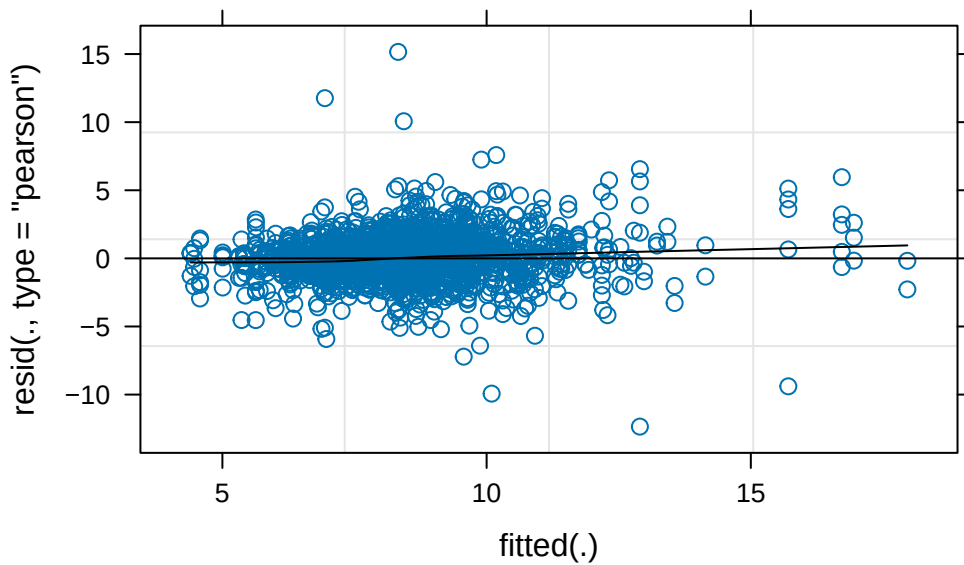
```
ci <- confint(mod1_outbed)
```

Computing profile confidence intervals ...

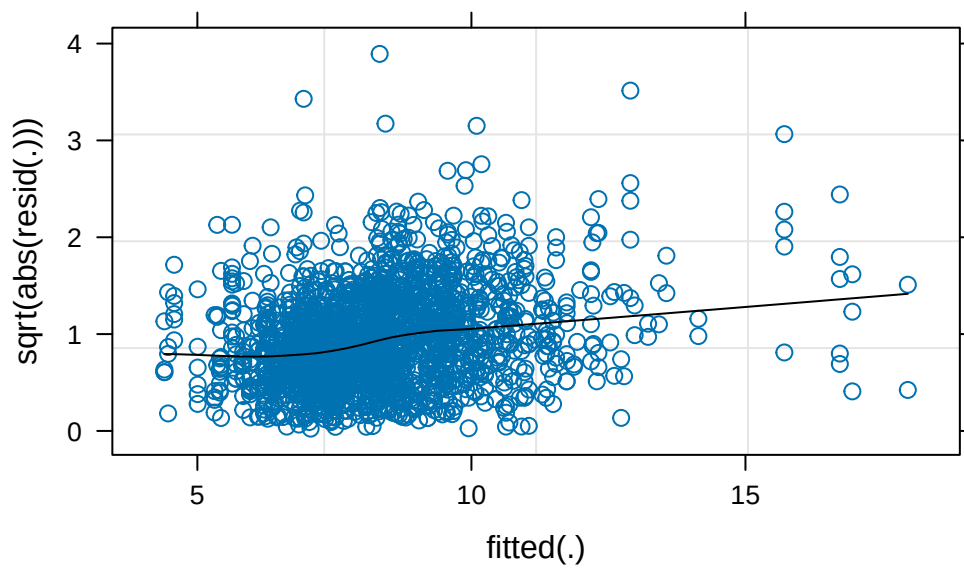
```
ci
```

	2.5 %	97.5 %
.sig01	1.49177265	1.80363000
.sigma	1.67919953	1.79251033
(Intercept)	7.51203976	9.70817400
pdsMF.v1male	-0.02748716	0.83961789
cg1.age.v1	-0.03278307	0.01071007
income.v1.10000	-0.10334362	0.03908259
cg1_edu_grp<=BS	-0.59748535	0.36201954
cg1_edu_grp>=MS	-0.54439843	1.10453969
I_frisat	1.07239976	1.40830479
z_COI_nat	-0.04665592	0.52716891

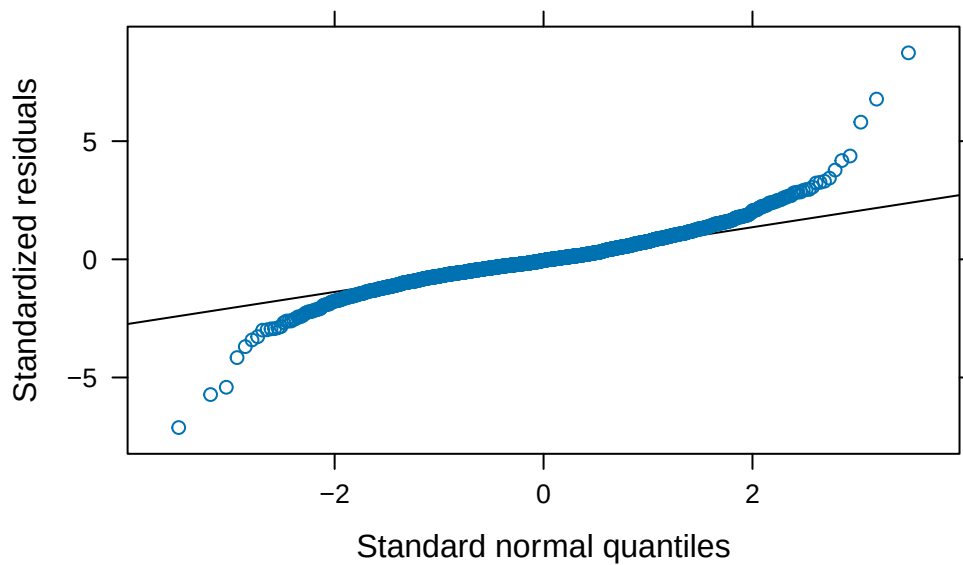
```
plot(mod1_outbed, type=c("p","smooth"), col.line=1)
```



```
plot(mod1_outbed,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod1_outbed)
```



```
mod1_outbed_table <- data.frame(
  predictor = attr(summary(mod1_outbed)$coef,"dimnames")[[1]],
  B = summary(mod1_outbed)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod1_outbed)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ", " ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_outbed)$CI_low,4),", ", " ",round(
  pval = summary(mod1_outbed)$coef[,5]
```

```
)

mod1_outbed_table[,c("B","beta","pval")] <-
round(mod1_outbed_table[,c("B","beta","pval")],4)
write.csv(mod1_outbed_table,file = "../bohnert_sleep_git/mod1_outbed_table.csv")

#Random effects
#within
sigma2 <- summary(mod1_outbed)$sigma^2
#between
tau00 <- attr(summary(mod1_outbed)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_outbed)$devcomp$dims["q"]
obs <- summary(mod1_outbed)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_outbed)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_outbed)[,"R2c"]
mod1_outbed_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_outbed_table_re
```

```

              sigma2    tau00        ICC    N  obs      R2marg    R2cond
(Intercept) 3.009453 2.760943 0.4784668 291 2094 0.06552546 0.5126405
```

```
#####
#Model 2: COI Sub-domains
#####
mod2_outbed <- lmer(act_outbed.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_
summary(mod2_outbed)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_outbed.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +
      cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat +      (1 | ID)
Data: datlong
```

REML criterion at convergence: 8850.7

Scaled residuals:

```

      Min       1Q   Median       3Q      Max
-7.1140 -0.4717 -0.0511  0.4518  8.7336
```

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.779	1.667
Residual		3.009	1.735

Number of obs: 2094, groups: ID, 291

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	8.51658	0.58369	282.03304	14.591	<2e-16 ***
pdsMF.v1male	0.40199	0.22444	279.02065	1.791	0.0744 .
cg1.age.v1	-0.01058	0.01125	280.11478	-0.940	0.3480
income.v1.10000	-0.02949	0.03725	277.57802	-0.792	0.4293
cg1_edu_grp<=BS	-0.11703	0.24813	283.88968	-0.472	0.6375
cg1_edu_grp>=MS	0.28998	0.42590	280.62101	0.681	0.4965
I_frisat	1.24053	0.08567	1813.90537	14.480	<2e-16 ***
z_ED_nat	0.12328	0.26769	285.25519	0.461	0.6455
z_HE_nat	0.18817	0.25360	280.07247	0.742	0.4587
z_SE_nat	0.01899	0.22339	281.93325	0.085	0.9323

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	pdMF.1	cg1...1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
pdsMF.v1mal	-0.111								
cg1.age.v1	-0.874	0.012							
inc.1.10000	-0.214	0.016	-0.017						
cg1_d_g<=BS	-0.254	-0.066	-0.003	-0.154					
cg1_d_g>=MS	-0.071	-0.142	-0.051	-0.297	0.467				
I_frisat	-0.038	0.003	-0.002	0.006	-0.004	-0.010			
z_ED_nat	0.014	0.056	-0.016	0.127	0.047	-0.008	0.002		
z_HE_nat	-0.203	-0.059	0.060	0.004	-0.032	0.019	0.003	-0.241	
z_SE_nat	0.213	-0.028	-0.039	-0.198	-0.067	-0.064	0.000	-0.711	-0.259

```
ci <- confint(mod2_outbed)
```

Computing profile confidence intervals ...

```
ci
```

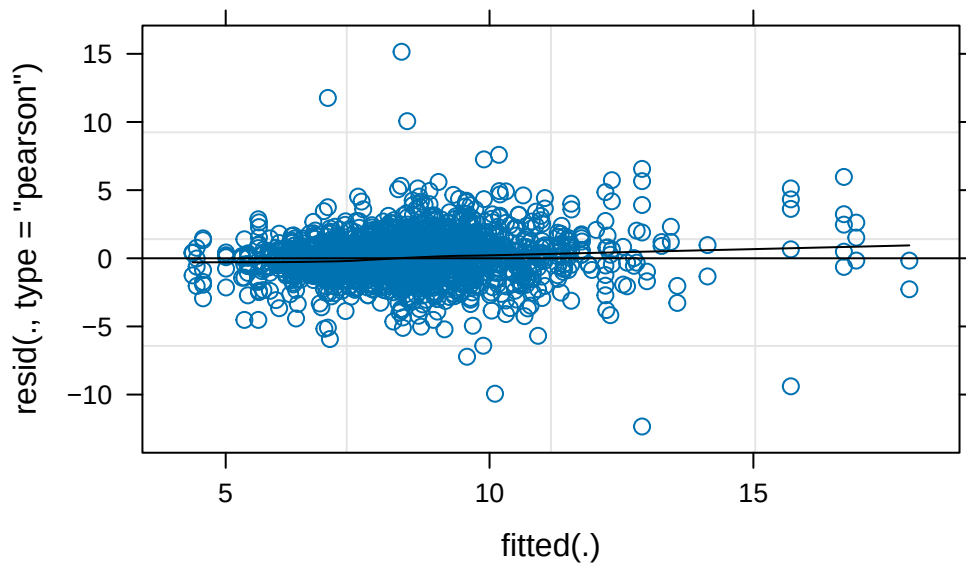
	2.5 %	97.5 %
.sig01	1.49059931	1.80220228
.sigma	1.67914825	1.79245122


```

(Intercept)      7.38662146  9.64633369
pdsMF.vmale     -0.03237113  0.83649769
cg1.age.v1      -0.03236033  0.01119464
income.v1.10000 -0.10158650  0.04262025
cg1_edu_grp<=BS -0.59714587  0.36351344
cg1_edu_grp>=MS -0.53412769  1.11467247
I_frisat        1.07261268  1.40850823
z_ED_nat        -0.39553698  0.64094019
z_HE_nat        -0.30285916  0.67890013
z_SE_nat        -0.41318209  0.45165630

```

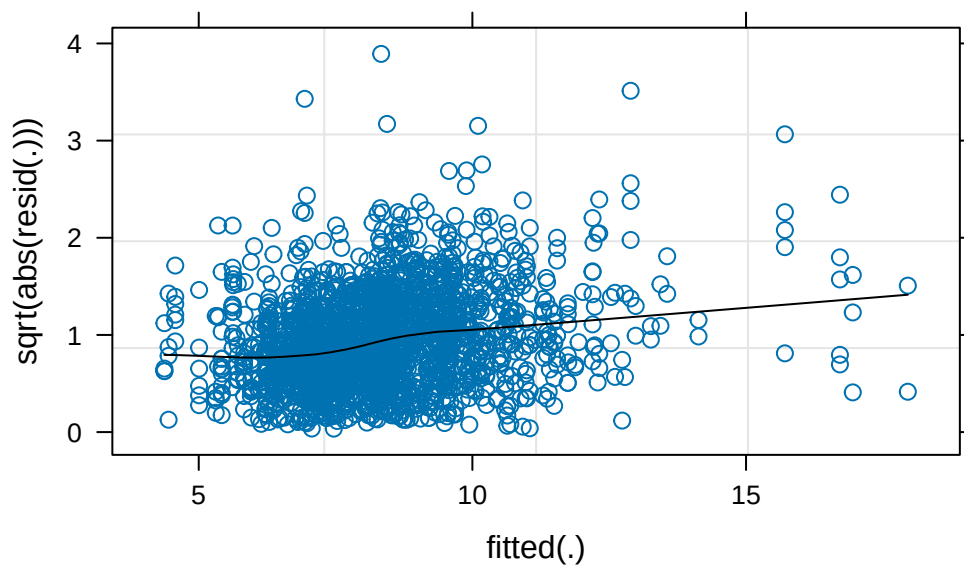
```
plot(mod2_outbed, type=c("p","smooth"), col.line=1)
```



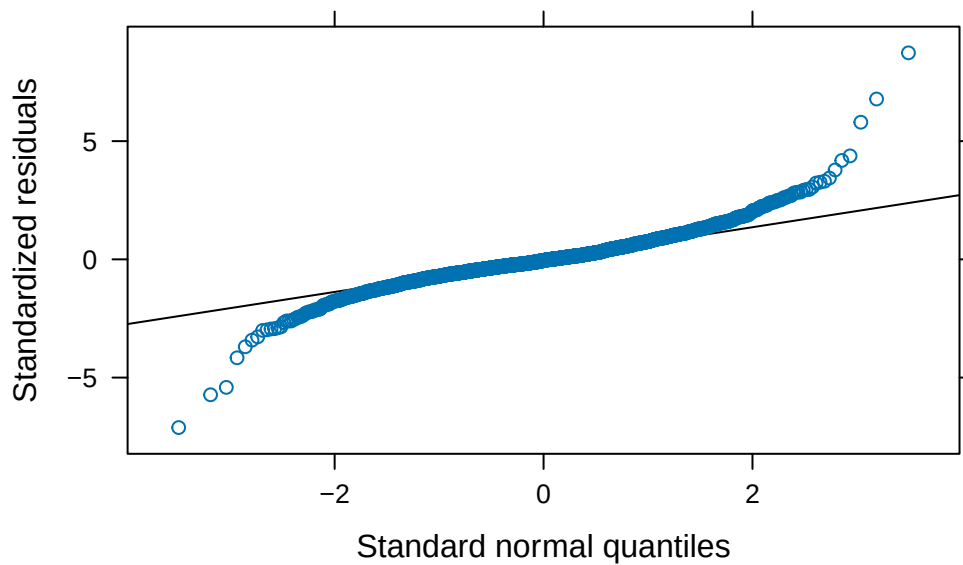
```

plot(mod2_outbed,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)

```



```
lattice::qqmath(mod2_outbed)
```



```
mod2_outbed_table <- data.frame(
  predictor = attr(summary(mod2_outbed)$coef,"dimnames")[[1]],
  B = summary(mod2_outbed)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod2_outbed)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ", " ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_outbed)$CI_low,4),",", " ",round(
  pval = summary(mod2_outbed)$coef[,5]
```

```
)

mod2_outbed_table[,c("B","beta","pval")] <-
round(mod2_outbed_table[,c("B","beta","pval")],4)
write.csv(mod2_outbed_table,file = "../bohnert_sleep_git/mod2_outbed_table.csv")

#Random effects
#within
sigma2 <- summary(mod2_outbed)$sigma^2
#between
tau00 <- attr(summary(mod2_outbed)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_outbed)$devcomp$dims["q"]
obs <- summary(mod2_outbed)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_outbed)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_outbed)[,"R2c"]
mod2_outbed_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_outbed_table_re
```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	3.009266	2.779257	0.4801324	291	2094	0.06601035	0.514449

```
#####
#Model 3: COI Sub-domains
#####
mod3_outbed <- lmer(act_outbed.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1_
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_outbed)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [

```

lmerModLmerTest]
Formula: act_outbed.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 +
      cg1_edu_grp + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
      z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
      z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat +
      (1 | ID)
Data: datlong

```

REML criterion at convergence: 7915.4

Scaled residuals:

Min	1Q	Median	3Q	Max
-6.8462	-0.5343	-0.0871	0.4971	8.0075

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.843	1.686
	Residual	3.454	1.859

Number of obs: 1821, groups: ID, 255

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	8.321877	0.687261	234.618287	12.109	<2e-16 ***
pdsMF.v1male	0.406433	0.249628	233.174910	1.628	0.105
cg1.age.v1	-0.009723	0.012154	234.171400	-0.800	0.425
income.v1.10000	-0.063340	0.041607	231.689543	-1.522	0.129
cg1_edu_grp<=BS	-0.145988	0.279750	237.109676	-0.522	0.602
cg1_edu_grp>=MS	0.375016	0.472333	234.792535	0.794	0.428
z_ED_EC_nat	-0.225218	0.138524	232.610578	-1.626	0.105
z_ED_EL_nat	0.495735	0.312466	236.901593	1.587	0.114
z_ED_ER_nat	-0.796100	0.540579	235.758079	-1.473	0.142
z_ED_SP_nat	0.200530	0.295790	235.452983	0.678	0.498
z_HE_EP_nat	1.832663	1.207660	233.969921	1.518	0.130
z_HE_HE_nat	0.805860	0.682990	236.068228	1.180	0.239
z_HE_HR_nat	0.021544	0.280768	233.642643	0.077	0.939
z_HE_SE_nat	0.049716	0.193922	237.749400	0.256	0.798
z_SE_EI_nat	0.675556	0.536549	235.816663	1.259	0.209
z_SE_EO_nat	-0.113192	0.358083	235.867489	-0.316	0.752
z_SE_ER_nat	-0.366078	0.366680	235.633891	-0.998	0.319
z_SE_HQ_nat	0.130852	0.338311	234.190697	0.387	0.699
z_SE_SR_nat	-0.302306	0.322251	234.072382	-0.938	0.349
z_SE_WL_nat	0.477768	0.387192	234.014331	1.234	0.218

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation matrix not shown by default, as $p = 20 > 12$.

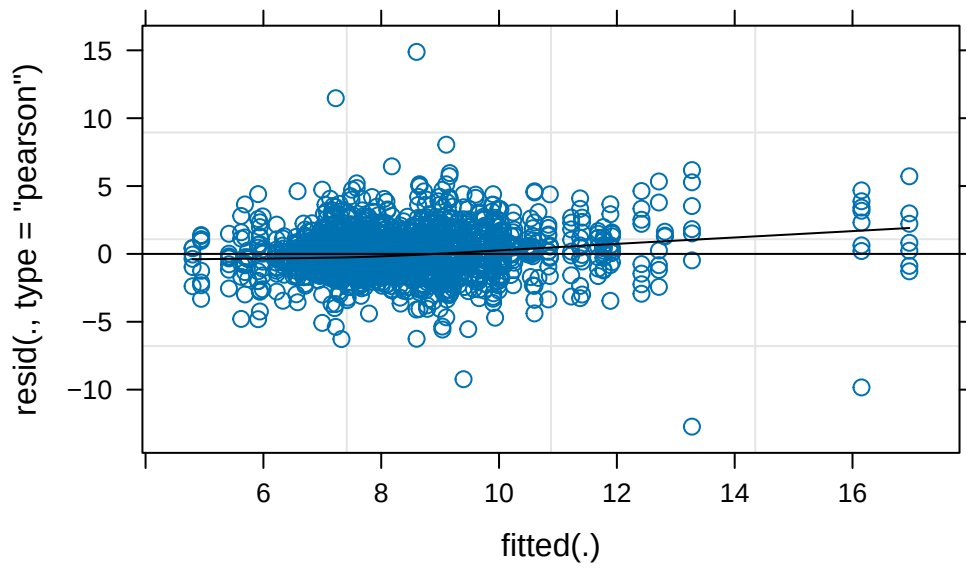
Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

```
confint(mod3_outbed)
```

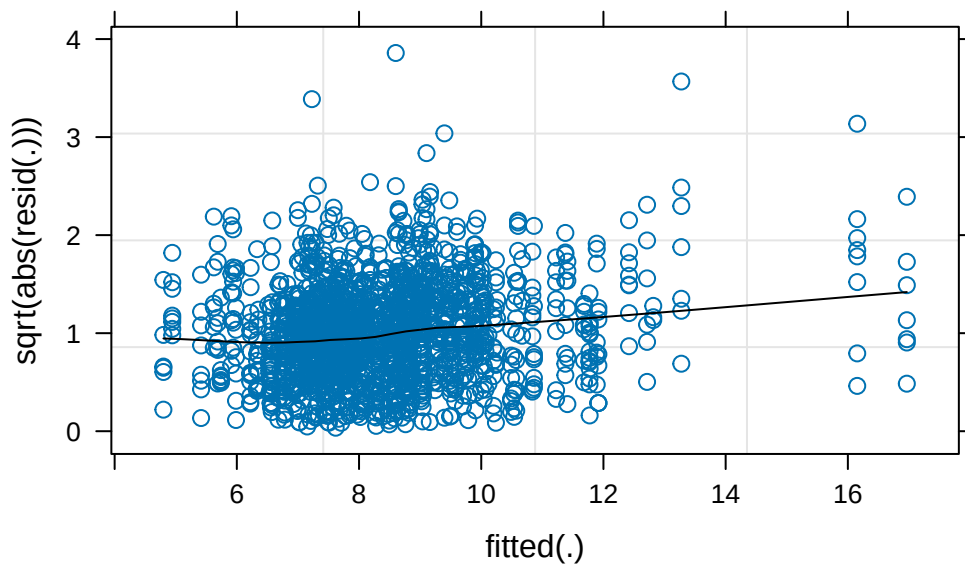
Computing profile confidence intervals ...

	2.5 %	97.5 %
.sig01	1.44908239	1.78465145
.sigma	1.79534924	1.92560140
(Intercept)	7.02473694	9.62063458
pdsMF.v1male	-0.06472117	0.87806646
cg1.age.v1	-0.03271122	0.01319460
income.v1.10000	-0.14185166	0.01527307
cg1_edu_grp<=BS	-0.67434134	0.38250910
cg1_edu_grp>=MS	-0.51630876	1.26780121
z_ED_EC_nat	-0.48648037	0.03667412
z_ED_EL_nat	-0.09626255	1.08420997
z_ED_ER_nat	-1.81837066	0.22366670
z_ED_SP_nat	-0.35973862	0.75761251
z_HE_EP_nat	-0.44866128	4.11267171
z_HE_HE_nat	-0.48512754	2.09491470
z_HE_HR_nat	-0.50986347	0.55059062
z_HE_SE_nat	-0.31614680	0.41649602
z_SE_EI_nat	-0.33612707	1.69070618
z_SE_EO_nat	-0.78888339	0.56378520
z_SE_ER_nat	-1.05999488	0.32514053
z_SE_HQ_nat	-0.50883473	0.76898868
z_SE_SR_nat	-0.90981464	0.30734871
z_SE_WL_nat	-0.25223326	1.21020739

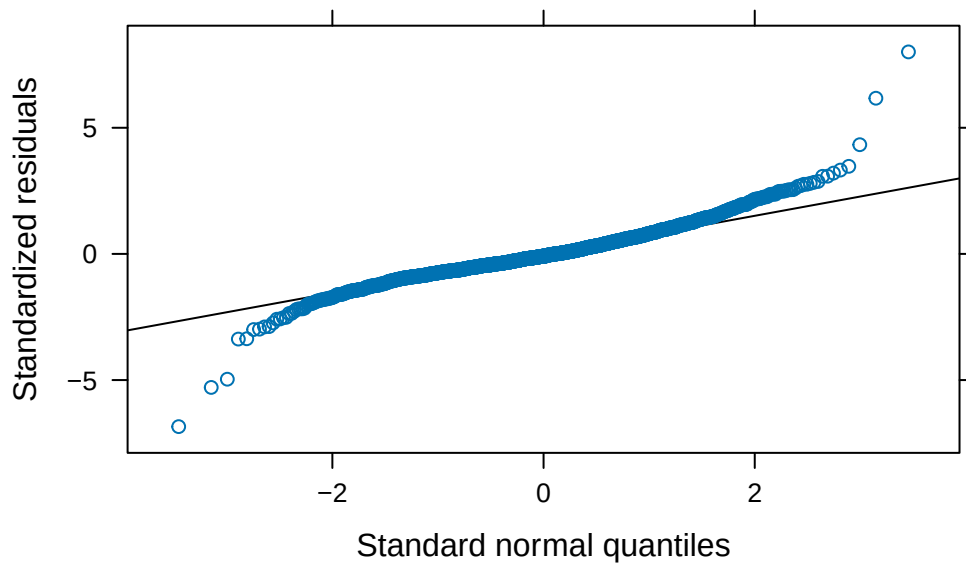
```
plot(mod3_outbed, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_outbed,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_outbed)
```



```

mod3_outbed_table <- data.frame(
  predictor = attr(summary(mod3_outbed)$coef,"dimnames")[[1]],
  B = summary(mod3_outbed)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod3_outbed)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_outbed)$CI_low,4),", ",round(effectsize::standardize_parameters(mod3_outbed)$CI_high,4),")"),
  pval = summary(mod3_outbed)$coef[,5]
)

mod3_outbed_table[,c("B","beta","pval")] <-
round(mod3_outbed_table[,c("B","beta","pval")],4)
write.csv(mod3_outbed_table,file = "../bohnert_sleep_git/mod3_outbed_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_outbed)$sigma^2
#between
tau00 <- attr(summary(mod3_outbed)$varcor$ID,"stddev")^2
ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_outbed)$devcomp$dims["q"]
obs <- summary(mod3_outbed)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_outbed)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_outbed)[,"R2c"]
mod3_outbed_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod3_outbed_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	3.454352	2.842608	0.4514255	255	1821	0.04413214	0.4756352

```
# #####
# #Model 4: Neighborhood Parent Report
# #####
# mod4_outbed <- lmer(act_outbed.time_numeric ~ pdsMF.v1 + cg1.age.v1 + income.v1.10000 + cg1
# summary(mod4_outbed)
# confint(mod4_outbed)
# plot(mod4_outbed, type=c("p","smooth"), col.line=1)
#
# plot(mod4_outbed,
#       sqrt(abs(resid(.)))~fitted(.),
#       type=c("p","smooth"), col.line=1)
#
# lattice::qqmath(mod4_outbed)
```